Younis Kernel Mixtures and Model-Score Estimation From the LEDH–OT–GenUT Filter to Corrected Mixtures

BayesFilter research note

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Abstract

A differentiable particle filter can compute the derivative of its own finite likelihood accurately while estimating the underlying model score poorly. This distinction explains both the progress and the negative results of the BayesFilter investigation of Younis and Sudderth's kernel mixtures. The original observation sidecar led to a complete kernelized-observation recursion and then to an all-components, raw-importance-weight resampling reference. The integrated derivatives pass their declared checks, but the calibrated resampling construction has higher model-score mean squared error than the canonical filter in both tested matrix-Gaussian scopes. We retain these results and, for models with regular transition densities, derive a different use of the same mixture ideas: construct evaluable proposals, retain the original transition and observation densities in the importance correction, and estimate the observed-data score by averaging complete-data scores over possible ancestors. This smoothing recursion does not apply as written to degenerate DSGE transitions. For those models we instead derive a disturbance-coordinate filter, its analytical state sensitivities, and an unbiased unnormalized likelihood–derivative pair. Propositions and proofs establish the importance correction, the particle induction, exact reference control variates, and conditional integration. A moving-support Gaussian example verifies the resulting score identity. Normalization can still introduce finite-particle bias, and long-horizon variance remains unresolved. Selective pathwise and importance-weight derivatives supply additional variance-reduction candidates.

The unbiasedness statement in the Younis–Sudderth construction has a more specific target. It is an importance-sampling identity for the gradient of a KDM expectation with the current proposal held fixed. When the integrand is a conditional log likelihood, the resulting quantity is an expected conditional log likelihood. It is not, in general, the log of the observed-data marginal likelihood, whose finite-particle score also contains a random normalization. The note documents the implemented finite programs, the reasons for each change in direction, the complete proposed recursions, and the experiments needed to distinguish improved model-score accuracy from implementation correctness. The new combination is a research proposal; its finite-particle bias and computational value remain to be measured.

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1 The model score and the finite filter

The scientific question is how to estimate the score of a state-space model accurately enough to support inference about its parameters. The intended BayesFilter LEDH calculation uses LEDH particle flow, PF–PF importance weights, entropy-regularized OT, Contract–E moment restoration, and GenUT higher-moment correction with two trust-region caps. These mechanisms define a particular finite calculation. They are useful starting points for a proposal, but they do not define what the exact model score must be.

This note grew out of the historical "Section 3.6" observation-sidecar document. That earlier note remains in the isolated `section-3-6-prior-weight-sidecar` worktree, under `docs/papers/section_3_6_prior_weight_sidecar/`. The present manuscript, `docs/papers/ledh_younis_kdm_score/ledh_younis_kdm_score.tex`, is its broader successor. It brings together the subsequent implementation, results, and the September 11 change in research question. The implementation history is tied to research commit `804616e3`; differences in the September comparison checkout, `5cc59cfa`, are identified in Section 13. These are historical snapshots, not the current integration checkout.

As of 14 October 2026, the separate controlled-SMC and Younis kernel-mixture replications have completed their declared, bounded released-source checks. This gives the project two tools to test: learned lookahead proposals from controlled SMC, and evaluable mixture proposals and mixture-expectation derivatives from KDM. It does not yet establish that combining either tool with LEDH improves the model score. The controlled-SMC replication is also distinct from the original iterated auxiliary particle filter, whose replication remains deferred. Section 7.2 explains the difference. While the LEDH tuning procedure is unavailable, the owner's temporary development baseline fixes transport regularization at $\epsilon = 1$ and the number of Sinkhorn iterations at 5. This is a fixed, untuned comparison setting, not a validated optimum for every model.

For fixed observations $y_{1:T}$ and model parameters $\theta \in \mathbb{R}^p$, write

$$\ell(\theta) = \log p_\theta(y_{1:T}), \qquad s(\theta) = \nabla_\theta \ell(\theta), \tag{1}$$

$$L_0^N(\theta;\xi) = \text{value of the executed finite filter}, \qquad S_0^N(\theta;\xi) = \nabla_\theta L_0^N(\theta;\xi). \tag{2}$$

Here N is the particle count and ξ is the fixed random design used by the finite program. In this note $\mathrm{d}F$ means the directional derivative $v^\mathsf{T}\nabla_\theta F$ for a declared direction v; s denotes the complete vector. Checking S_0^N against finite differences of L_0^N tests the implementation of that derivative. Comparing it with s tests its accuracy as a model-score estimator. A third calculation, developed in Section 9, estimates s through posterior expectations. It need not equal S_0^N for the same particle realization.

For a fixed dataset, define the conditional bias and covariance of the finite-program score by

$$b_N(\theta;y) = \mathbb{E}_\xi[S_0^N \mid y] - s(\theta), \qquad V_N(\theta;y) = \mathrm{Cov}_\xi(S_0^N \mid y). \tag{3}$$

When the second moments exist, its squared Euclidean error has expectation $\|b_N\|^2 + \operatorname{tr} V_N$. An individual discrepancy from Kalman does not identify which term is responsible. Averaging errors over different observation paths answers another question again: conditional biases can cancel across datasets. Repeated filter randomizations within each dataset are needed to distinguish conditional bias from Monte Carlo variance.

Even an unbiased likelihood estimate does not generally give an unbiased log-likelihood score. Consider the positive estimator $\widehat{Z}(\theta) = \theta + \varepsilon$, where $\Pr(\varepsilon = a) = \Pr(\varepsilon = -a) = 1/2$ and $\theta > a > 0$. Its mean is $Z(\theta) = \theta$, and its derivative is exactly one. Yet

$$\mathbb{E}\left[\partial_\theta \log \widehat{Z}\right] = \frac{1}{2}\left(\frac{1}{\theta + a} + \frac{1}{\theta - a}\right) = \frac{\theta}{\theta^2 - a^2} \neq \frac{1}{\theta} = \partial_\theta \log Z. \tag{4}$$

At $\theta = 2$ and $a = 1/2$, the two scores are $8/15$ and $1/2$. The bias is created by the random denominator, despite correct differentiation and an unbiased numerator. This example does not say that every finite-particle score is biased in every model. It shows why the two correctness statements cannot be substituted for a score-accuracy argument.

There are consequently several possible repairs. Missing derivatives of the state, covariance, initial law, or sampling distribution require an implementation or estimator correction. A reset or kernel that changes the filtering measure requires a correction to that approximation or an explicit new target. A valid but noisy integral requires better integration. The development below separates these explanations before selecting the next method. A zero-mean control variate addresses the last problem; it does not remove an arbitrary bias already present in its baseline.

The existing implementation names remain useful for identifying what ran. `ATOM-FINITE` denotes L_0^N; `KDM-FINITE` denotes a positive-bandwidth observation program; `RESKDM-IWSG-FINITE` denotes the post-Contract–E resampling program with raw anchored ratios; and `RESKDM-SN-FINITE` denotes its historical, prematurely normalized variant. `MODEL-IS` denotes an importance calculation whose numerator retains the original model factors. These are program identities, not levels of scientific validity. In particular, a correct `MODEL-IS` update does not by itself make an additional deterministic reset unbiased, and a different finite derivative is acceptable only under its actual name.

The LEDH spelling follows the Li–Coates particle-flow construction [Li and Coates, 2017]; the differentiable-transport context is Corenflos et al. [2021]. The new investigation now includes alternative model-score estimators explicitly. It no longer rules them out merely because they differentiate a different finite scalar.

2 The executed atom program

Let $x_{t,i}^-$ and $w_{t,i}^-$ denote the cloud and normalized weights immediately before the observation update. The finite program can be written schematically as

$$x_{t,i}^- = \Phi_{t,\theta}(x_{t-1,i}^+, \xi_t), \tag{5}$$

$$x_{t,i}^{\mathrm{out}} = F_{t,\theta}^{\mathrm{LEDH}}(x_{t,i}^-; \xi_t), \tag{6}$$

$$a_{t,i} = \log w_{t,i}^- + \log p_\theta(x_{t,i}^{\mathrm{out}} \mid x_{t-1,i}^+) + \log g_\theta(y_t \mid x_{t,i}^{\mathrm{out}}) + \log |J_{t,i}| - \log q_{t,\theta}(x_{t,i}^- \mid x_{t-1,i}^+, y_t), \tag{7}$$

$$Z_t = \sum_{i=1}^N \exp(a_{t,i}), \tag{8}$$

$$w_{t,i}^+ = \exp(a_{t,i})/Z_t, \tag{9}$$

$$x_t^+ = R_\theta(x_t^{\mathrm{out}}, w_t^+; \xi_t), \tag{10}$$

$$L_0^N(\theta; \xi) = \sum_{t=1}^T \log Z_t, \qquad S_0^N(\theta; \xi) = \nabla_\theta L_0^N(\theta; \xi). \tag{11}$$

Here $x_{t,i}^{\mathrm{out}}$ is the post-flow state, $J_{t,i}$ is the forward flow Jacobian, and $q_{t,\theta}$ is the pre-flow proposal density. Φ includes the declared transition and UKF covariance lifecycle, while F^{LEDH} is the flow map; the displayed terms make the PF–PF correction explicit. In a bootstrap/no-flow special case, the proposal and transition terms cancel and $a_{t,i} = \log w_{t,i}^- + \log g_\theta(y_t \mid x_{t,i}^-)$. The reset R includes the Sinkhorn/OT map, Contract–E moments, GenUT corrections, and the dual-cap trust-region map. This form matches the canonical score route, whose weight logits contain transition, observation, log-determinant, and proposal terms. No such term may be silently dropped from the value or its tangent.

For the auxiliary sidecar it is useful to expose the same code path as a prior-observation factorization. Define

$$b_{t,i} = \log w_{t,i}^{-} + \log p_\theta(x_{t,i}^{\mathrm{out}} \mid x_{t-1,i}^{+}) + \log |J_{t,i}| - \log q_{t,\theta}(x_{t,i}^{-} \mid x_{t-1,i}^{+}, y_t), \tag{12}$$

$$B_t = \log \sum_i \exp(b_{t,i}), \qquad \bar{w}_{t,i} = \exp(b_{t,i} - B_t), \tag{13}$$

$$a_{t,i} = b_{t,i} + \log g_\theta(y_t \mid x_{t,i}^{\mathrm{out}}), \qquad Z_t = \sum_i \exp(a_{t,i}). \tag{14}$$

The implementation records $\bar{w}_t$ as `prior_observation_weights`; these are the weights that may be used with a new observation functional. The posterior weights in (10) are w_t^+ and must not be reused as $\bar{w}_t$.

The derivative in (2) is a total derivative. Thus a tangent through x_t^-, w_t^-, the observation factor, the OT map, the Contract–E residual design, the higher moments, and the dual caps is included whenever that dependence is part of the declared finite scalar. Holding a cloud fixed and differentiating only the observation factor is a different partial derivative.

The executable analytical endpoint evaluates one arbitrary parameter direction per call. For a scalar parameter this number is the score. For $p > 1$, a full score vector requires p calls with the model tangent set to the coordinate directions (or another spanning basis); each call is still a total derivative through the entire finite recurrence. The Phase 4A and Phase 4B score-quality fixtures have one parameter. The later adapter audit exercises coordinate and mixed directions for five models through the two-step endpoints. Those checks cover the derivative components, but do not establish a scalable batch implementation of the full score vector.

The research snapshot exposes the following analytical function and its legacy row-mapped batch wrapper:

- `canonical_value_and_analytical_score` `bayesfilter/highdim/ledh_canonical_score_tf.py`;
- the batch wrapper `canonical_batch_value_score` `bayesfilter/highdim/ledh_canonical_batch_tf.py`.
  It accepts both the historical `substeps` keyword and the authority-aligned `flow_substeps` alias, and rejects conflicting values.

The separate lower-level GenUT candidate/diagnostic is:

- `finite_value_score` `bayesfilter/highdim/cubature_genut_filter.py`.

Calling it is not evidence that the LEDH PF–PF flow and Contract–E reset were executed. The registry mismatch found during the audit has been repaired: the registry now names the exported `canonical_value_and_analytical_score`, and the governance test resolves every registered entry point to a callable. This closes discovery only; it does not implement KDM or prove numerical score validity. The existing Section 3.6 prior-weight sidecar is diagnostic work from an isolated worktree; it is not an integrated route in this checkout and is not a Younis IWSG implementation through the reset. The analytical function defaults to `reset_policy=none`, a diagnostic slice. A canonical baseline invocation must explicitly select `reset_policy=contract_e` and the owner-registered dual-cap and trust-region settings.

3 What the Younis construction supplies

The mixture resampling and fixed-proposal importance gradient below follow Younis and Sudderth [2023, Section 4, equations (14)–(15)]. We distinguish that construction from its later integration with BayesFilter resets and scores.

Younis and Sudderth begin with a weighted empirical cloud and replace its Dirac representation by a regularized mixture

$$\mu_h^N(\mathrm{d}z) = \sum_{i=1}^{N} w_i k_{B_i}(z - x_i)\,\mathrm{d}z, \qquad m_\theta(z) = \sum_{i=1}^{N} w_i(\theta) k_{B_i(\theta)}(z - x_i(\theta)). \tag{15}$$

For their mixture-density particle filter, this mixture replaces discrete resampling. With a common Gaussian bandwidth one may generate an anchor sample by

$$I_j \sim \mathrm{Cat}(w_1, \ldots, w_N), \qquad z_j = x_{I_j} + L\varepsilon_j, \qquad \varepsilon_j \sim N(0, I), \quad LL^{\mathsf{T}} = B. \tag{16}$$

The 2024 construction stratifies the uniforms used for the categorical draw. The label I_j generates the sample, but the differentiable density is the marginal sum in (15); it is not the density of the selected component. Thus every component can contribute to the gradient at every sample location. This is the source of the N-by-N work in a gradient-bearing resampling step.

Their IWSG construction fixes a proposal at the current parameter value. With

$$q_0(z) = m_{\theta_0}(z), \qquad \rho_j(\theta) = \frac{m_\theta(z_j)}{q_0(z_j)}, \qquad z_j \sim q_0, \tag{17}$$

the finite estimator of a mixture expectation and its derivative are

$$\widehat{I}(\theta) = \frac{1}{M}\sum_{j=1}^{M} \rho_j(\theta)\phi_\theta(z_j), \tag{18}$$

$$\mathrm{d}\widehat{I} = \frac{1}{M}\sum_{j=1}^{M} \left\{\mathrm{d}\rho_j\, \phi_\theta(z_j) + \rho_j\, \mathrm{d}\phi_\theta(z_j)\right\}. \tag{19}$$

At $\theta = \theta_0$, $\rho_j = 1$. The locations z_j are held fixed in this differentiation; parameter dependence is represented by the mixture density ratio. This is not the total JVP of an inverse-CDF draw or of a parameter-dependent component index. Equations (17)– (19) establish the derivative of a mixture expectation. They do not by themselves establish the derivative of an arbitrary nonlinear operation that couples the complete set $(z_1, \ldots, z_M)$. For such an operation H_θ, ordinary importance sampling on the joint sample would instead use

$$R_M(\theta) = \prod_{j=1}^{M} \rho_j(\theta), \qquad \mathrm{d}\{R_M H_\theta\}\big|_{\theta_0} = \mathrm{d}H_{\theta_0} + H_{\theta_0}\sum_{j=1}^{M} \mathrm{d}\log m_\theta(z_j)\big|_{\theta_0}. \tag{20}$$

This joint ratio is exact for an expectation under independent mixture draws, but its variance can grow rapidly with M. Feeding individual IWSG weights through a nonlinear OT reset defines a finite weighted-particle algorithm; it is not silently promoted to the joint identity (20).

The ratio in (17) is a raw importance weight. Replacing it by the self-normalized ratio

$$\bar{\rho}_j(\theta) = \frac{\rho_j(\theta)}{\sum_l \rho_l(\theta)} \tag{21}$$

changes the finite estimator. At the anchor, $\mathrm{d}\log\rho_j = \mathrm{d}\log m_\theta(z_j)$, whereas

$$\mathrm{d}\log\bar{\rho}_j = \mathrm{d}\log m_\theta(z_j) - \frac{1}{M}\sum_l \mathrm{d}\log m_\theta(z_l). \tag{22}$$

The second expression has been centered by a random sample average and is not Equation (15) of Younis and Sudderth. Their released MDPF code confirms the same ordering: it constructs the unnormalized gradient-injection factor $\exp\{\log m - \mathrm{stop}(\log m)\}$, multiplies that factor by the next measurement and dynamics weights, and only then normalizes the posterior weights [Younis, 2023]. The source code's small additive numerical constant is an engineering detail and is not part of the mathematical route defined here.

For a Gaussian component, let $u_i = z - x_i$. Its complete directional differential is

$$\mathrm{d}\log k_{B_i}(u_i) = u_i^{\mathsf{T}} B_i^{-1}\mathrm{d}x_i + \frac{1}{2}\left[u_i^{\mathsf{T}} B_i^{-1}(\mathrm{d}B_i)B_i^{-1}u_i - \mathrm{tr}(B_i^{-1}\mathrm{d}B_i)\right]. \tag{23}$$

Therefore

$$\mathrm{d}m_\theta(z) = \sum_i \left[(\mathrm{d}w_i)k_{B_i}(u_i) + w_i k_{B_i}(u_i)\mathrm{d}\log k_{B_i}(u_i)\right]. \tag{24}$$

Equation (24) is the term that a KDM score path must carry for every cloud, weight, or bandwidth dependence declared to vary with θ. A bandwidth or cloud may instead be a genuinely fixed numerical design, in which case its zero tangent is part of the scalar's definition. Stopping a dependence that the declared scalar retains is a partial derivative; declaring a quantity fixed before defining the scalar is not.

3.1 The precise scope of IWSG unbiasedness

The word "unbiased" in the IWSG argument refers to a particular expectation and a particular differentiation convention. It should not be read as a claim that every finite-particle log-likelihood score is unbiased. Let m_θ be a positive mixture density with respect to a fixed reference measure and let $F_\theta(z)$ be a scalar downstream integrand. Define

$$\mathcal{J}(\theta) = \int F_\theta(z) m_\theta(z)\,\mathrm{d}z. \tag{25}$$

Fix an anchor θ_0, condition on the current incoming cloud, and draw $z_j \sim q_0$ with $q_0(z) = m_{\theta_0}(z)$. The proposal q_0 is detached during the local differentiation. Assume that the support is common in a neighbourhood of θ_0 and that differentiation may pass through the integral. The fixed-proposal estimator is

$$\widehat{G}_M = \frac{1}{M}\sum_{j=1}^{M}\left[\nabla_\theta F_\theta(z_j)|_{\theta_0} + F_{\theta_0}(z_j)\,\nabla_\theta \log m_\theta(z_j)|_{\theta_0}\right]. \tag{26}$$

Proposition 1 (IWSG expectation-gradient identity)**.** *Under the conditions above, conditional on the current cloud,*

$$\mathbb{E}_{q_0}[\widehat{G}_M] = \nabla_\theta \mathcal{J}(\theta)|_{\theta_0}. \tag{27}$$

The same statement holds for a general proposal q_0 when each summand is multiplied by $m_{\theta_0}(z_j)/q_0(z_j)$.

Proof. At the anchor, $q_0 = m_{\theta_0}$. Therefore the expectation of one summand in (26) is

$$\begin{aligned}
&\int q_0(z) \left[\nabla_\theta F_\theta(z)|_{\theta_0} + F_{\theta_0}(z)\, \nabla_\theta \log m_\theta(z)|_{\theta_0} \right] \mathrm{d}z \\
&= \int \left[\nabla_\theta F_\theta(z)|_{\theta_0}\, m_{\theta_0}(z) + F_{\theta_0}(z)\, \nabla_\theta m_\theta(z)|_{\theta_0} \right] \mathrm{d}z \\
&= \nabla_\theta \int F_\theta(z) m_\theta(z)\, \mathrm{d}z \Big|_{\theta_0}.
\end{aligned}$$

The average has the same expectation. For a general proposal, inserting the ratio m_{θ_0}/q_0 gives the same integral. The denominator is fixed because it is the law used to generate the current sample; differentiating it would define a different estimator rather than the local derivative in (25). □

The proposition is a conditional statement: it is made after the current mixture parameters and incoming cloud have been fixed. If those quantities are random, the identity can be averaged again by the tower property, but only for the expectation objective actually defined by the complete recursion. Any upstream dependence of the cloud that the declared objective retains must still be differentiated. Holding the proposal density fixed is a local sampling convention; it does not authorize dropping total derivatives elsewhere in the filter.

The distinction becomes visible when the integrand is a log likelihood. If

$$F_\theta(z) = \log g_\theta(y \mid z), \qquad \mathcal{J}_{\mathrm{cond}}(\theta) = \mathbb{E}_{m_\theta}\left[\log g_\theta(y \mid Z)\right], \tag{28}$$

then Proposition 1 estimates the gradient of the expected conditional log likelihood. In the simple case where m_θ is the prior, the corresponding observed-data marginal log likelihood is

$$\ell_{\mathrm{marg}}(\theta) = \log \int g_\theta(y \mid z) m_\theta(z)\, \mathrm{d}z. \tag{29}$$

The logarithm is outside the integral in (29), whereas it is inside the expectation in (28). Consequently,

$$\mathbb{E}_{m_\theta}[\log g_\theta(y \mid Z)] \neq \log \mathbb{E}_{m_\theta}[g_\theta(y \mid Z)] \tag{30}$$

except in special cases. Jensen's inequality gives the direction in the ordinary prior case, but it does not turn the two objectives into the same score. A finite particle average makes the distinction concrete:

$$\frac{1}{M} \sum_j \log g_\theta(y \mid z_j) \neq \log \left(\frac{1}{M} \sum_j g_\theta(y \mid z_j) \right). \tag{31}$$

For the generative state-space score, write the model normalizer as

$$Z(\theta) = \int u_\theta(z)\, \mathrm{d}z, \qquad \ell(\theta) = \log Z(\theta), \tag{32}$$

where u_θ contains the initial, transition, and observation factors in the declared model. With a fixed proposal q_0, importance sampling can give unbiased estimators of the unnormalized value and derivative,

$$\begin{aligned}
\widehat{Z}_M &= \frac{1}{M} \sum_j \frac{u_{\theta_0}(z_j)}{q_0(z_j)}, & \mathbb{E}[\widehat{Z}_M] &= Z(\theta_0), \\
\widehat{D}_M &= \frac{1}{M} \sum_j \frac{\nabla_\theta u_\theta(z_j)|_{\theta_0}}{q_0(z_j)}, & \mathbb{E}[\widehat{D}_M] &= \nabla_\theta Z(\theta)|_{\theta_0}.
\end{aligned} \tag{33}$$

The natural finite score is nevertheless

$$\widehat{S}_M = \frac{\widehat{D}_M}{\widehat{Z}_M}, \tag{34}$$

and generally

$$\mathbb{E}[\widehat{S}_M] \neq \frac{\mathbb{E}[\widehat{D}_M]}{\mathbb{E}[\widehat{Z}_M]} = \nabla_\theta \log Z(\theta_0). \tag{35}$$

This is a normalization and ratio issue, not a failure of the local IWSG identity. An expectation of a complete-data log density can equal a marginal log likelihood only under an additional exact-posterior identity, for example

$$\log p_\theta(y) = \mathbb{E}_{p_\theta(z|y)} \left[\log p_\theta(y, z) - \log p_\theta(z \mid y)\right]. \tag{36}$$

A KDM mixture used as an approximate filter or proposal is not automatically that posterior, and $F = \log g$ omits the second, entropy term in (36).

For a sequential particle filter, each local KDM/IWSG step can therefore be mathematically correct for its declared mixture expectation while the full finite program still differs from the exact marginal score. Repeated normalization, nonlinear resampling or transport, and a changed positive-bandwidth filtering measure all affect the complete target. The `RESKDM-IWSG-FINITE` route defined below has a valid total derivative of its own finite scalar; the source IWSG result does not identify that scalar with $\nabla_\theta \log p_\theta(y_{1:T})$ without a separate target and finite-particle analysis.

The source papers establish this mixture-gradient construction for learned, discriminative state-density objectives. In particular, the 2023 MDPF uses the mixture both for posterior-state loss evaluation and for resampling; the 2024 paper again gives the marginalized ratio and separates resampling and posterior bandwidths. Neither paper derives the observed-data score of a generative state-space model, an LEDH flow, an OT reset, GenUT moments, or dual-cap constraints. Their complexity discussion makes the computational boundary explicit: evaluating the mixture gradient at every resampled location creates all-pairs interactions. The papers' linear-cost inference path sets forward resampling weights to uniform and does not compute these gradients. It cannot be transplanted to a score path without a new approximation and error analysis.

4 Why positive bandwidth is a target change

The atom cloud is

$$\mu_0^N(\mathrm{d}z) = \sum_i w_i \delta_{x_i}(\mathrm{d}z). \tag{37}$$

For a test function ϕ,

$$\int \phi(z)\mu_h^N(\mathrm{d}z) = \sum_i w_i \int \phi(x_i + u)K_{B_i}(\mathrm{d}u), \tag{38}$$

which is not generally $\sum_i w_i \phi(x_i)$. This gives the following simple boundary.

Proposition 2. *Suppose the kernel weights are nonnegative and sum to one, and each kernel is centered with finite covariance B_i. If $w_i > 0$ and $B_i \neq 0$ for at least one component, the atom and kernel measures differ. Their parameter derivatives therefore require separate justification; a positive-bandwidth calculation is not, by its construction, the derivative of the atom program.*

Proof. Take $\phi(z) = \|z\|^2$. Centering eliminates the cross term in $\|x_i + u\|^2$, so the difference between the two expectations is $\sum_i w_i \operatorname{tr}(B_i) > 0$. Unequal parameterized expectations can nevertheless have equal derivatives at particular points, or even differ by a constant. Thus this measure argument

proves a change of functional, not a universal pointwise inequality of scores. Equality of the scores would need an additional argument for the actual parameterized functional. □

A limiting theorem as bandwidth tends to zero would concern convergence; it would not make two finite-bandwidth programs exactly equal.

The case $B = 0$ is an explicit atom branch, not a request to evaluate a singular Gaussian density. A zero-bandwidth parity test must therefore not form a Cholesky factor or an inverse of $B = 0$ and then infer that the result is the atom program.

For a linear-Gaussian observation model, a Gaussian state kernel gives the useful exact convolution

$$\overline{g}_\theta(y \mid x, B) = \int g_\theta(y \mid x + u) N(u; 0, B)\, \mathrm{d}u = N\Big(y; C_\theta x,\, R_\theta + C_\theta B C_\theta^{\mathsf{T}}\Big). \tag{39}$$

Write

$$S_i = R_\theta + C_\theta B_i C_\theta^{\mathsf{T}}, \qquad \mu_i = C_\theta x_i, \qquad r_i = S_i^{-1}(y - \mu_i). \tag{40}$$

For fixed data y, the total directional differential used below is

$$\mathrm{d}\mu_i = (\mathrm{d}C)x_i + C\,\mathrm{d}x_i, \tag{41}$$

$$\mathrm{d}S_i = \mathrm{d}R + (\mathrm{d}C)B_i C^{\mathsf{T}} + C(\mathrm{d}B_i)C^{\mathsf{T}} + CB_i(\mathrm{d}C)^{\mathsf{T}}, \tag{42}$$

$$\mathrm{d}\log \overline{g}_i = r_i^{\mathsf{T}}\mathrm{d}\mu_i + \frac{1}{2}\left\{r_i^{\mathsf{T}}(\mathrm{d}S_i)r_i - \mathrm{tr}(S_i^{-1}\mathrm{d}S_i)\right\}. \tag{43}$$

Thus state, observation-map, observation-covariance, and bandwidth tangents are all required when those quantities vary. Positive semidefinite B_i is permitted here, including rank-deficient B_i, because only the strictly positive definite observation covariance S_i is factored. This statement does not assert an ambient Lebesgue density for a singular state kernel. The corresponding observation increment

$$\overline{Z}_t = \sum_i w_{t,i}^- \overline{g}_\theta(y_t \mid x_{t,i}^-, B_{t,i}), \qquad \overline{L}_h^N = \sum_t \log \overline{Z}_t \tag{44}$$

is a well-defined new finite program. It equals the atom increment only at zero bandwidth or in special accidental cases. If it is fed into the subsequent reset, it defines an even larger new program.

The same distinction applies to a KDM placed after the reset. Replacing x_t^+ by a draw from (15) changes the input to the next transition and therefore changes all later normalizers. The change may be scientifically useful, but it is not an implementation detail.

5 From Section 3.6 to the integrated experiments

The earliest question was whether an inexpensive smooth observation functional could provide useful score information alongside the GenUT filter. The Section 3.6 sidecar was designed as a separate deterministic proposal force, with the canonical value retained for endpoint correction. That was a legitimate mechanics experiment, but it was not a construction of the canonical total score and it never supplied a known-mean control variate.

Its first repair concerned observation timing. Let w_i^- be normalized weights before the current observation, g_i its likelihood at particle i, and $Z = \sum_i w_i^- g_i$. The posterior weights are $w_i^+ = w_i^- g_i / Z$. The historical capture hook ran before reset but after this observation update. Reusing its w_i^+ in a second likelihood average therefore gave

$$\widetilde{Z} = \sum_i w_i^+ g_i = \frac{\sum_i w_i^- g_i^2}{Z}, \qquad \widetilde{Z} - Z = \frac{\mathrm{Var}_{w^-}(g)}{Z}. \tag{45}$$

For two equally weighted particles with likelihoods 1 and 3, the intended increment is 2, while the reused posterior weights give 2.5. The word "pre-reset" did not make the captured weights pre-observation. The repaired sidecar used the actual prior observation weights and differentiated its declared observation covariance dependence, while holding the captured cloud, weights, and kernel fixed. It remained a conditional derivative.

The September 3 pilot used one-step diagonal LGSSMs: $(d, N, T) = (1, 4, 1)$ and $(3, 6, 1)$. Four calibration paths and eight streams per scope selected the grid boundary $\rho = 0.8$; eight disjoint validation paths and eight streams per scope supplied the holdout comparison. The recorded relative path-level squared-bias changes were 6.6082 with interval $[-0.2825, 19.4854]$ and 0.2209 with interval $[-0.9342, 1.5096]$. Both intervals crossed zero, and the recorded stream-MCSE ratios, ranging from 0.1756 to 0.4709, failed the threshold 0.10. These tiny CPU/non-XLA diagnostics supplied no ranking or bias-reduction evidence. Separate reversibility and volume checks passed at about machine precision, but proposal mechanics did not establish score accuracy, HMC mixing, or posterior correctness. The sidecar remained isolated and unpromoted.

The next stages sought to remove the conditional-derivative restriction. Phases 1–3 implemented complete mixture differentials, a fixed-chart support reference, and an auxiliary functional constructed from the actual canonical trace. Phase 4A then fed kernelized observation weights and their tangents through the complete reset. Phase 4B added the source paper's mixture resampling mechanism after Contract–E and carried the resulting weights and covariance marks into later time steps. These changes made the full feedback testable. They also defined new finite programs; no residual correction made them equal to the original atom program.

The chronology therefore does not support a claim that a control variate was completed and then simplified. The executed work moved from an auxiliary conditional functional to separately named integrated experiments. This was useful for determining whether those mechanisms improved model-score error. It did not solve the original target-preserving control-variate problem. The negative results below are the reason to reconsider the insertion point and the score estimator together.

5.1 Phase 4A: integrated kernelized observation weighting

The first integrated experiment replaces only the point observation factor in (14). With the ordinary pre-observation PF–PF logit $b_{t,i}$, define

$$a^h_{t,i} = b_{t,i} + \log \overline{g}_\theta(y_t \mid x^{\mathrm{out}}_{t,i}, B_{t,i}), \tag{46}$$

$$Z^h_t = \sum_i \exp(a^h_{t,i}), \qquad \alpha^h_{t,i} = \exp(a^h_{t,i})/Z^h_t, \tag{47}$$

$$L^N_{\mathrm{obsKDM}} = \sum_t \log Z^h_t, \qquad S^N_{\mathrm{obsKDM}} = \mathrm{d}L^N_{\mathrm{obsKDM}}. \tag{48}$$

Its one-step tangent is

$$\mathrm{d}\log Z^h_t = \sum_i \alpha^h_{t,i} \left(\mathrm{d}b_{t,i} + \mathrm{d}\log \overline{g}_{t,i}\right), \qquad \mathrm{d}\alpha^h_{t,i} = \alpha^h_{t,i} \left[\mathrm{d}a^h_{t,i} - \mathrm{d}\log Z^h_t\right]. \tag{49}$$

The executed program passes $(x^{\mathrm{out}}_t, \mathrm{d}x^{\mathrm{out}}_t, \alpha^h_t, \mathrm{d}\alpha^h_t)$ into the existing Contract–E reset and the existing GenUT/dual-cap correction. Their output and tangent feed the next time step. Equations (43) and (49) therefore define a total derivative of this complete finite program, not a conditional observation derivative.

The reset also carries the particle-local UKF covariance with the same target-by-source transport used for the state. If A_{ri} is the normalized Sinkhorn transport from source particle i to reset particle r, then the covariance recursion and its total tangent are

$$P'_r = \sum_i A_{ri} P_i, \qquad \mathrm{d}P'_r = \sum_i (\mathrm{d}A_{ri}) P_i + \sum_i A_{ri} \mathrm{d}P_i. \tag{50}$$

The carried P'_r, not the old source ordering, is consumed by the next UKF prediction. Here P_i denotes a particle-local conditional covariance; no between-cloud scatter term is added. This same-transport carry is the declared Contract–E extension of categorical triple resampling. A previous implementation omitted it at the analytical endpoint; that call-chain bug was repaired and is now covered by endpoint and two-step tests.

This program is deliberately called *kernelized observation weighting*. It is not the complete Bayesian update of a Gaussian mixture: it does not propagate the conditional posterior mean and covariance of every mixture component. It is consequently not a full Younis mixture-density particle filter. Positive bandwidth is labelled `KDM-FINITE`; the explicit $B = 0, \mathrm{d}B = 0$ branch is labelled `ATOM-FINITE` and must reproduce the canonical trajectory.

A second insertion point represents the post-reset cloud by a continuous mixture before the next transition. It defines the distinct `RESKDM-IWSG-FINITE` scalar and is derived and implemented separately in the next subsection; it is not part of the Phase 4A observation-weighting scalar.

The first paired experiment now answers the narrow Phase 4A question for one small linear–Gaussian cell. With $N = 32$, $T = 5$, 20 calibration paths, and 100 disjoint validation paths, calibration selected $\rho = 0.8$ in $B = \rho^2 Q$. The atom score MSE was 3.3366 and the kernelized-observation MSE was 3.4980, a relative change of -4.84% when positive values denote improvement. The paired 95% bootstrap interval for the MSE difference was $[-0.223, 0.597]$, so the two scores are not statistically ranked. Inspection of all prespecified bandwidths on this now-consumed holdout showed no positive point improvement; $\rho = 1.6$ was worse with interval $[0.189, 2.399]$.

The mechanism is visible in the score errors. From $\rho = 0$ to $\rho = 0.8$, their sample standard deviation decreased from 1.616 to 1.580, but their mean moved from -0.868 to -1.014. The modest variance reduction was more than offset by additional bias. This rejects Phase 4A for promotion in this scope. It does not test the complete Younis resampling operation described next, and it does not license tuning on the inspected validation paths.

For a finite IWSG normalizer,

$$\widehat{Z}_h(\theta) = \frac{1}{M} \sum_j \rho_j(\theta) g_\theta(y_t \mid z_j), \tag{51}$$

$$\mathrm{d} \log \widehat{Z}_h = \frac{\sum_j \rho_j g_j \left(\mathrm{d} \log \rho_j + \mathrm{d} \log g_j \right)}{\sum_j \rho_j g_j}. \tag{52}$$

Equation (52) is a derivative of the KDM importance-sampling program. It is not an identity with the atom program.

5.2 Phase 4B: a complete mixture-resampling reference

The source operation can now be stated without an ancestry ambiguity. Let $(c_{t,i}, \alpha_{t,i}, P_{t,i})$ denote the locations, normalized weights, and particle-local UKF covariance marks presented to a continuous resampler. If the KDM replaces Contract–E, these are the weighted post-flow particles. In the combined BayesFilter reference selected below, they are the Contract–E output, its carried

covariances, and uniform weights $\alpha_{t,i} = 1/N$ with $\mathrm{d}\alpha_{t,i} = 0$. In either case define the marginalized resampling density

$$m_{t,\theta}(z) = \sum_{i=1}^{N} \alpha_{t,i}(\theta) k_{B_t(\theta)}(z - c_{t,i}(\theta)). \tag{53}$$

The first reference uses the source paper's common Gaussian bandwidth and an ordinary full-rank linear–Gaussian fixture. The complete sequential route is not called source-faithful because its Contract–E placement and covariance marks are BayesFilter extensions. A DSGE version must replace this density by the chart density in Section 12.

At an anchor θ_0, stratified categorical uniforms and Gaussian noises generate fixed locations $z_{t,j}$ from m_{t,θ_0}. Component labels are recorded to replay the draw, but are marginalized in every density evaluation. For a nearby θ, define

$$r_{t,j}(\theta) = \frac{m_{t,\theta}(z_{t,j})}{m_{t,\theta_0}(z_{t,j})}, \qquad \eta_{t,j}(\theta) = \frac{1}{N} r_{t,j}(\theta), \tag{54}$$

$$\gamma_{t,ji}(\theta) = \frac{\alpha_{t,i} k_{B_t}(z_{t,j} - c_{t,i})}{m_{t,\theta}(z_{t,j})}. \tag{55}$$

At the anchor, $r_{t,j} = 1$ and $\eta_{t,j} = 1/N$. The locations are fixed in the IWSG differential, so $\mathrm{d}z_{t,j} = 0$, while

$$\mathrm{d}\log r_{t,j} = \sum_i \gamma_{t,ji} \left\{ \mathrm{d}\log \alpha_{t,i} + \mathrm{d}\log k_{B_t}(z_{t,j} - c_{t,i}) \right\}, \tag{56}$$

$$\mathrm{d}\log \eta_{t,j} = \mathrm{d}\log r_{t,j}, \qquad \mathrm{d}\eta_{t,j} = \eta_{t,j} \mathrm{d}\log r_{t,j}. \tag{57}$$

Equations (23) and (57) include the cloud, weight, and bandwidth terms. Evaluating all $\gamma_{t,ji}$ is the required N^2 operation. The sum of the $\eta_{t,j}$ tangents is generally nonzero for a finite sample; forcing it to zero would replace the raw IWSG correction by (21).

Younis's particles carry locations and weights, not the UKF covariance used by the LEDH implementation. The following mark transport is therefore an explicit BayesFilter extension:

$$\widetilde{P}_{t,j} = \sum_i \gamma_{t,ji} P_{t,i}, \qquad \mathrm{d}\widetilde{P}_{t,j} = \sum_i (\mathrm{d}\gamma_{t,ji}) P_{t,i} + \sum_i \gamma_{t,ji} \mathrm{d}P_{t,i}, \tag{58}$$

$$\mathrm{d}\gamma_{t,ji} = \gamma_{t,ji} \left\{ \mathrm{d}\log \alpha_{t,i} + \mathrm{d}\log k_{t,ji} - \mathrm{d}\log m_{t,\theta}(z_{t,j}) \right\}. \tag{59}$$

No between-component scatter is added: the kernel perturbation has already been realized in $z_{t,j}$, and $P_{t,i}$ remains the local UKF algorithmic mark. This choice is testable, but it is not a result from the source paper. The prespecified comparator instead carries the covariance mark of the fixed anchor label,

$$\widetilde{P}^{\mathrm{label}}_{t,j} = P_{t,I_{t,j}}, \qquad \mathrm{d}\widetilde{P}^{\mathrm{label}}_{t,j} = \mathrm{d}P_{t,I_{t,j}}. \tag{60}$$

The label has no tangent. Both policies retain the same marginalized mixture density and raw IWSG weights; only the downstream UKF mark differs. Neither mark policy is supplied by Younis and Sudderth, so their comparison diagnoses a BayesFilter extension rather than source fidelity.

That comparison is uninformative when all component marks and their tangents are equal. If $P_{t,i} = P_t$ and $\mathrm{d}P_{t,i} = \mathrm{d}P_t$, then $\sum_i \gamma_{t,ji} = 1$ and $\sum_i \mathrm{d}\gamma_{t,ji} = 0$ give

$$\widetilde{P}_{t,j} = P_t, \qquad \mathrm{d}\widetilde{P}_{t,j} = \Big(\sum_i \mathrm{d}\gamma_{t,ji} \Big) P_t + \Big(\sum_i \gamma_{t,ji} \Big) \mathrm{d}P_t = \mathrm{d}P_t. \tag{61}$$

The fixed-label policy gives the same result. Linear Gaussian prediction and update preserve equal initial covariance marks, and normalized Contract–E transport preserves their equality. The matrix experiment below therefore cannot select between these policies; floating-point differences in their calibration scores have no scientific meaning. Heterogeneous-mark tests are needed to check the distinction, and nonlinear evidence is needed to choose between them for a nonlinear filter.

There are now two possible placements, and they must not share a name. Replacing Contract–E by mixture resampling gives the source MDPF placement but removes the requested Contract–E, GenUT, and dual-cap reset. Retaining Contract–E and then applying (54) before the next transition executes both mechanisms. The latter is the declared Phase 4B BayesFilter reference. Its fixed locations, raw IWSG weights, and covariance marks become the parents of the next transition; all later PF–PF, flow-Jacobian, reset, and cap terms are then differentiated normally.

To make the finite scalar explicit, let $h_{t+1,j}(\theta)$ contain the full next-step PF–PF multiplier for child j: transition density, observation density, flow Jacobian, and proposal denominator. The post-resampling increment is

$$Z_{t+1}^{\mathrm{IWSG}}(\theta) = \sum_{j=1}^{N} \eta_{t,j}(\theta) h_{t+1,j}(\theta) = \frac{1}{N} \sum_{j=1}^{N} r_{t,j}(\theta) h_{t+1,j}(\theta). \tag{62}$$

The posterior weights are normalized only after the product $\eta_{t,j} h_{t+1,j}$ is formed. This preserves the raw importance-sampling normalizer in (62). Call the resulting scalar

$$L^N_{\mathrm{resKDM,IWSG}}(\theta; \xi, z^0, q^0) = \sum_{t=1}^{T} \log Z_t^{\mathrm{IWSG}}, \qquad S^N_{\mathrm{resKDM,IWSG}} = \mathrm{d}L^N_{\mathrm{resKDM,IWSG}}. \tag{63}$$

This is the combined `RESKDM-IWSG-FINITE` program. It is neither L_0^N nor the Younis MDPF alone. A two-pass evaluation is required: an anchor pass generates every sequential $z_{t,j}$ and denominator $m_{t,\theta_0}(z_{t,j})$; a replay pass evaluates the same fixed sample bank and its total tangent for other θ. Re-anchoring inside an HMC trajectory would change the finite energy and is forbidden.

The raw IWSG weights are part of this new finite scalar, not auxiliary metadata. In particular, if $\omega_{t+1,j}$ denotes the incoming weight used by the next PF–PF step, then

$$\log \omega_{t+1,j} = \log r_{t,j} - \log N, \qquad \mathrm{d} \log \omega_{t+1,j} = \mathrm{d} \log r_{t,j}. \tag{64}$$

The next normalizer must consume both quantities. Resetting the numerical weights to $1/N$ while retaining only the sampled locations, or retaining the weight values while discarding their tangent, removes the IWSG mechanism.

Prematurely normalizing the ratios defines a different scalar. Because the subsequent posterior softmax removes common scale, the state and posterior weight recurrences agree, but the accumulated normalizer does not. With the same samples and covariance-mark policy,

$$L^N_{\mathrm{resKDM,SN}} = L^N_{\mathrm{resKDM,IWSG}} - \sum_{t=1}^{T-1} \log \left\{ \frac{1}{N} \sum_{j=1}^{N} r_{t,j} \right\}. \tag{65}$$

The first Phase 4B implementation computed this `RESKDM-SN-FINITE` scalar. Its finite-difference agreement was correct for that scalar but did not validate the intended raw-IWSG score. The repaired route implements (64); an executable regression requires a generally nonzero sum of raw weight tangents and rejects the centered tangent in (22) as its incoming score.

There are also two distinct meanings of "unbiased" that must not be conflated. Equations (17)–(19) give the ordinary importance-sampling identity for a single mixture expectation. For a nonlinear function of the complete resampled cloud, the exact joint identity is (20). The sequential use of raw individual weights in (64) follows the IWSG gradient-injection ordering, while using their scale in a generative evidence normalizer is a BayesFilter extension. Its analytical derivative is the total derivative of $L^N_{\mathrm{resKDM,IWSG}}$ as defined here, but neither the source paper nor the present derivation proves that it is an unbiased estimator of $\nabla_\theta \log p_\theta(y_{1:T})$, of the expected stochastic-filter loss, or of S_0^N. Those are empirical or additional-theory questions.

The zero-bandwidth limit requires separating the locations from their IWSG derivative. In this combined route, the component weights are exactly $1/N$ and the jth stratified position is $(j-1+u_j)/N$, with $0 \le u_j < 1$. Consequently the selected label is $I_j = j$: each component is selected once. The earlier statement that this particular route tends to repeated component locations was wrong. With $B = h^2 Q$ and $L_Q L_Q^\mathsf{T} = Q$, its anchor sample is $z_j = c_j + hL_Q\varepsilon_j$, which tends to c_j.

This location limit does not establish score parity. For distinct centres, fixed positive-definite Q, and a parameter-independent bandwidth scale h, the selected component dominates the mixture as $h \to 0$. At the anchor, its leading location contribution to the IWSG differential is

$$\mathrm{d}\log m_\theta(z_j)\big|_{\theta_0} \simeq \frac{1}{h}\varepsilon_j^\mathsf{T} L_Q^{-1}\mathrm{d}c_j. \tag{66}$$

Thus the fixed-sample tangent can grow while the forward locations converge. Repeated centres require a separate mixture-limit analysis. At $B = 0$ the ambient Gaussian density is undefined, so Phase 4B does not implement a zero-bandwidth score branch equal to `ATOM-FINITE`. Its checks are instead anchor identity, complete-mixture density, analytical-versus-finite-difference tangents, sequential replay, and comparison with both the canonical score and the Kalman score.

The exact all-components reference costs $O(N^2)$ per mixture-gradient evaluation and $O(TN^2)$ for a sequential gradient. Sparse neighbors, random features, and low-rank approximations would be new algorithms; they are not permitted in the reference and require separate error analyses if considered later.

5.3 The matrix Gaussian score comparison

The first score-quality campaign used a two-state model. Its dynamics are $x_t = A(\theta)x_{t-1} + \epsilon_t$, and its measurements satisfy $y_t = Hx_t + \nu_t$. The matrices are

$$A(\theta) = \begin{bmatrix} 0 & 0.12 \\ -0.08 & 0.10 \end{bmatrix} + \theta \begin{bmatrix} 1 & 0 \\ 0 & 0.7 \end{bmatrix}, \qquad \theta_0 = 0.72, \tag{67}$$

$$H = \begin{bmatrix} 1 & 0.25 \\ -0.15 & 0.9 \end{bmatrix}, \qquad Q = \begin{bmatrix} 0.12 & 0.025 \\ 0.025 & 0.09 \end{bmatrix}, \tag{68}$$

$$R = \begin{bmatrix} 0.22 & 0.035 \\ 0.035 & 0.18 \end{bmatrix}, \qquad P_0 = \begin{bmatrix} 0.8 & 0.12 \\ 0.12 & 0.6 \end{bmatrix}. \tag{69}$$

Here $x_{-1} \sim N(0, P_0)$ and the first observation follows one transition. At θ_0, $A(\theta_0)$ is stable. An independent analytical Kalman recursion differentiates its prediction, innovation covariance, gain, posterior mean, and posterior covariance with $\mathrm{d}A = \mathrm{diag}(1, 0.7)$. The first particle scopes were $(N, T) = (32, 5)$ and $(64, 20)$. The smaller scope ran first; both scopes completed within the total compute budget. This matrix fixture replaces the earlier scalar proposal, which was wrong for testing the full pairwise correction because that correction is identically absent in one dimension.

Each scope used 40 calibration paths, 30 separate power-pilot paths, and 500 untouched validation paths. The bandwidths selected from the declared grid were $\rho = 0.8$ and 0.4, respectively. Relative to the exact Kalman score, the validation result was

(N, T)	Atom MSE	KDM MSE	KDM minus atom, 95% interval
$(32, 5)$	3.94210	4.45250	[0.07106, 0.94986]
$(64, 20)$	3.86260	10.13463	[5.10083, 7.55881]

The intervals use 4,000 paired path bootstrap resamples. They support higher MSE for the calibrated KDM candidate in each tested scope. Both frozen low/high absolute-score groups also had higher KDM point MSE than the atom and observation-KDM comparators; the high-score small-scope interval alone crossed zero. The simpler bootstrap particle comparator had larger MSE, which does not rescue the loss to the unmodified filter.

The primary criterion required an interval demonstrating at least ten percent MSE reduction. Neither scope passed it. The pilots requested 8,900 and 4,160 paths for the specified twenty-percent improvement alternative, so the 500-path cap also left both scopes underpowered for that design. The negative result is therefore a rejection of these calibrated candidates, not a completed search over KDM methods. Their numerical controls remain unpromoted hypotheses, and changing N and T together does not isolate either effect. In the smaller scope, a slight observed variance reduction was offset by greater squared bias; in the larger scope both increased. These bias/variance decompositions are descriptive. Small bandwidth was especially unstable: the calibration MSE at $\rho = 0.025$ reached 1.63×10^9 and 9.18×10^{35}. This is consistent with the small-bandwidth tangent in (66), not evidence that a small value perturbation gives a small score perturbation.

The bootstrap entry differentiates the finite program along its realized, fixed ancestor labels. It checks that program's derivative; it is not the ancestor-averaged Fisher-score baseline developed in Section 9. The canonical numerical controls were frozen campaign comparators and were not independently optimized for each scope. The results therefore compare the tested configurations. Since both N and T change between the two scopes, the contrast does not isolate horizon dependence or particle-number dependence. The reported subgroup intervals are pointwise, not simultaneous.

The raw-IWSG log-value shift relative to the atom program averaged -0.02449 and -0.02377 in the two scopes, with mean squared shifts 0.07284 and 0.04023. Small changes in a scalar do not imply small changes in its derivative. These descriptive value differences therefore do not rescue the failed score comparisons.

5.4 Why the next direction changes

The tested all-IWSG candidate is rejected for the two matrix-Gaussian scopes. Its derivative agrees with its declared finite program, yet its model-score MSE is higher. In the smaller scope, the descriptive error mean and variance changed from -1.08143 and 2.77816 for the canonical score to -1.32100 and 2.71290 for Phase 4B. In the larger scope they changed from -0.98767 and 2.89289 to -1.57710 and 7.66270. The second scope therefore shows both more squared error in the mean and more variation across paths. These across-path decompositions are descriptive; they do not separately estimate the conditional bias at each fixed dataset.

The result is not adequately explained by saying that the KDM derivative is different from the atom derivative. Both are candidates for approximating the exact model score, and a different estimator could have smaller error. The more useful question is why the new approximation should be closer to the model. Adding Gaussian observation covariance changes the observation factor. Adding a second reset changes future particles. Evaluating narrow kernels with fixed-location

importance gradients can amplify noise. None of these operations, by itself, corrects the original model-score bias.

The earlier restriction to the executed finite derivative made unrelated score recursions appear inadmissible. That restriction protected the meaning of the canonical API, but it was too narrow as a research objective. A Fisher-score estimator can be a valid candidate for the underlying model without being the derivative of the existing OT-reset scalar. It must be named and validated accordingly. Likewise, a centered control variate can reduce variance without removing the bias of its baseline. The distinction changes what an experiment must measure; it does not relax the mathematics.

One historical cost argument also needs correction. The Section 3.6 memo grouped PaRIS with inherently quadratic ancestor calculations. The exact ancestor sum is quadratic, but PaRIS replaces that sum with sampled backward ancestors and has an expected linear-cost result under stated mixing and acceptance assumptions [Olsson and Westerborn, 2014, Section 3.1, Algorithm 2; Section 3.2.4, Theorem 10]. Those assumptions do not hold automatically in every repository model, particularly when transition densities are singular or backward acceptance becomes small. "Always quadratic" was therefore an incorrect cost argument. Correcting that argument does not remove the structural obstruction from degenerate transitions: sampling backward ancestors cannot create transitions that the model forbids. A comparison with ancestor-averaged methods is a question for models whose transition and backward-conditioning assumptions hold.

For those regular-transition models, a reference comparison assigns different jobs to three components. An evaluable mixture proposes particles. The original model densities specify their importance correction. A posterior-score recursion averages over the ancestor uncertainty that remains. Selective pathwise and importance-weight gradients are an additional way to reduce variance when mixture expectations must be differentiated. This combination has not been tested here. Its components have explicit roles in the model-score calculation, but that does not make it a general solution for the degenerate DSGE target. The support obstruction in Section 9.1 must be resolved before proposing a score recursion for that target.

6 Mixture proposals with the original model correction

There is a continuous mixture already present in a particle filter whenever the physical transition has a continuous density. Let $\{x_{t-1}^j, W_{t-1}^j\}_{j=1}^N$ approximate the filtering distribution after observation $t-1$, with $\sum_j W_{t-1}^j = 1$. Propagating that measure through the actual transition density gives

$$a_t(x) = \sum_{j=1}^{N} W_{t-1}^j f_\theta(x \mid x_{t-1}^j), \qquad u_t(x) = g_\theta(y_t \mid x) a_t(x). \tag{70}$$

The first density predicts the state; the second incorporates the current observation but is not normalized. The transition's process covariance is part of the model. It is not an artificial bandwidth added to make an atom cloud differentiable. When f_θ is Gaussian, a_t is already a Gaussian mixture, so Younis-style mixture calculus can be studied on this physical predictive distribution directly.

Choose an evaluable proposal q_t whose support contains that of u_t, and draw $x_t^i \sim q_t$. The correct importance calculation is

$$v_t^i = \frac{u_t(x_t^i)}{q_t(x_t^i)}, \qquad \widehat{z}_t = \frac{1}{N} \sum_i v_t^i, \qquad W_t^i = \frac{v_t^i}{\sum_k v_t^k}. \tag{71}$$

Conditionally on the incoming cloud and a proposal fixed before these draws,

$$\mathbb{E}_{q_t}[\widehat{z}_t \mid \mathcal{F}_{t-1}] = \int \frac{u_t(x)}{q_t(x)} q_t(x)\,\mathrm{d}x = \int u_t(x)\,\mathrm{d}x. \tag{72}$$

Here $\mathcal{F}_{t-1}$ includes the previous weighted particles and any proposal design fixed from them and the observed data. A kernel can change where particles are drawn without changing the numerator in this identity. The cancellation fails if the denominator is merely the density of the selected component rather than the actual marginal proposal, or if the proposal is refitted to the same draws without a justified adaptive weighting scheme.

This is the useful connection to the marginal particle filter. Lai et al. [2022, Algorithm 2, Theorem 2, equations (9)–(12)] integrate out the ancestor index and evaluate full transition and proposal mixtures. Their estimator-coupling argument establishes an unbiased likelihood normalizer for the specified recursion. It also explains a local Rao–Blackwell variance reduction; the paper explicitly does not claim that the complete MPF always has lower variance than every corresponding SMC run. Its variational objective, equation (13), is $\mathbb{E}\log\widehat{Z}$, which is not $\log Z$. Unbiased differentiation of that objective would not settle the model-score question in Section 1.

The associated author implementation evaluates the transition mixture with `reduce_logsumexp`, adds the observation log density, and subtracts the proposal log density [Lai, 2022]. These operations support (71). They do not establish compatibility with BayesFilter's analytical derivative, initialization, GPU/XLA, or tuning requirements. The proposed generative mixture route is not implemented by merely calling the existing observation-KDM endpoint.

The distinction between numerator and proposal also determines how much of the current LEDH machinery can be reused. Its flow, UKF information, GenUT moments, and OT couplings may supply proposal centers, covariances, or deterministic transformations. The numerator must still use the intended incoming filtering weights and the actual transition and observation densities. Replacing those weights by uniform reset weights and those states by uncorrected reset locations changes a_t. Equation (72) then certifies an integral of that changed approximation, not the original model recursion. A deterministic transformation may be used as a proposal only when its induced density, including every necessary Jacobian, is known.

A defensive mixture can protect proposal support when a guided component misses a plausible mode. For a fixed $0 < \eta < 1$, let

$$q_t(x) = (1-\eta)q_t^{\mathrm{guided}}(x) + \eta a_t(x). \tag{73}$$

Since $q_t \geq \eta a_t$, the incremental weight satisfies $v_t(x) \leq g_\theta(y_t \mid x)/\eta$. A bounded observation likelihood therefore gives a bounded incremental weight. It does not bound every score derivative, and the mixing fraction remains a scope-specific proposal choice to calibrate. An arbitrary positive fraction is not justified as a default.

6.1 An analytically adapted Gaussian example

The first implementation can be checked in a model where the best local proposal is available exactly. Suppose $f_\theta(x \mid x_{t-1}^j) = \mathcal{N}(x; m_j, Q)$ and $g_\theta(y_t \mid x) = \mathcal{N}(y_t; Hx, R)$, with Q and R positive definite. In the matrix-LGSSM test, $m_j = A(\theta)x_{t-1}^j$. Multiplying the two Gaussian densities gives

$$f_\theta(x \mid x_{t-1}^j) g_\theta(y_t \mid x) = c_j\,\mathcal{N}(x; \widetilde{m}_j, \widetilde{Q}), \tag{74}$$

where completing the square, or conditioning a joint Gaussian, yields

$$V = HQH^{\mathsf{T}} + R, \qquad c_j = \mathcal{N}(y_t; Hm_j, V), \tag{75}$$

$$K = QH^{\mathsf{T}}V^{-1}, \qquad \widetilde{m}_j = m_j + K(y_t - Hm_j), \tag{76}$$

$$\widetilde{Q} = Q - QH^{\mathsf{T}}V^{-1}HQ. \tag{77}$$

The component evidence c_j tells us how plausible ancestor j is after the new observation. Its conditional Gaussian moves the proposal toward that observation. Thus the normalized predictive update is the mixture

$$q_t^*(x) = \sum_j \frac{W_{t-1}^j c_j}{z_t^*} \mathcal{N}(x; \widetilde{m}_j, \widetilde{Q}), \qquad z_t^* = \sum_j W_{t-1}^j c_j. \tag{78}$$

Substituting $q_t^* = u_t / z_t^*$ into (71) makes every incremental weight equal to z_t^*. The current-step proposal variance vanishes conditionally on the incoming cloud. Earlier particle approximation remains, so this is not an exact finite-particle Kalman filter. The example isolates the improvement available from conditional integration and proposal adaptation before any extra kernel smoothing is introduced. In particular, Q here is the physical process covariance, not the carried UKF mark or an added kernel bandwidth.

6.2 Retained ancestry and a flow proposal

The same model correction can be formulated on a joint ancestor–state space. This is useful when the flow depends on the selected ancestor and its inverse cannot be marginalized cheaply. The distinction determines the correct numerator as well as the denominator; the following existing `MODEL-IS` derivation is retained for that reason.

There is one direct KDM use that leaves the generative model factors in the importance ratio, but it is a BayesFilter extension rather than the Younis MDPF. The treatment of ancestry must be fixed before writing the ratio. If an ancestor A is retained, draw it with a frozen probability $r_{t,0}(a)$, then draw a pre-flow state from a complete conditional mixture $q_{t,0}^{\mathrm{KDM}}(x_0 \mid a, y_t)$. Apply the ancestor-specific LEDH map $x_1 = F_{t,\theta,a}(x_0)$, with forward Jacobian $J_{t,a}$. The induced joint proposal density is

$$q_t(j, x_1) = r_{t,0}(j) q_{t,0}^{\mathrm{KDM}}(x_0 \mid j, y_t) \, |\det J_{t,j}(x_0)|^{-1}, \qquad x_0 = F_{t,\theta,j}^{-1}(x_1). \tag{79}$$

The corresponding unnormalised weight is

$$\widetilde{w}_t = \frac{w_{t-1,J}^- f_\theta(x_1 \mid x_{t-1,J}^+) g_\theta(y_t \mid x_1) |\det J_{t,J}(x_0)|}{r_{t,0}(J) q_{t,0}^{\mathrm{KDM}}(x_0 \mid J, y_t)}. \tag{80}$$

Here "complete" means the sum over every kernel component within the conditional mixture for the retained ancestor, not the kernel of the component that generated x_0. If ancestry is marginalized instead, the numerator must also be marginalized,

$$\widetilde{w}_t(x_0) = \frac{g_\theta(y_t \mid x_1) |\det J_t(x_0)| \sum_a w_{t-1,a}^- f_\theta(x_1 \mid x_{t-1,a}^+)}{q_{t,0}^{\mathrm{KDM}}(x_0)}. \tag{81}$$

This second expression is valid only when the flow map does not depend on an ancestor that has been marginalized. Mixing a marginalized denominator with the conditional numerator in (80) is wrong.

An auxiliary-particle lookahead may guide $r_{t,0}$, but its approximate predictive likelihood cannot replace either the exact transition factor f_θ or the observation factor g_θ. Under the usual support, invertibility, and differentiation-under-the-integral conditions, the sample average $\widehat{Z}_t$ and its derivative are unbiased for the one-step particle predictive integral conditional on the incoming cloud. The ratio $\mathrm{d}\widehat{Z}_t/\widehat{Z}_t$ is generally biased for its logarithmic derivative. Moreover, the deterministic Contract–E reset changes the next incoming cloud, so this one-step identity is not a proof that the complete finite-horizon algorithm is an unbiased likelihood or score estimator.

Following the fixed-denominator principle of Younis IWSG, this conditional LEDH extension freezes $r_{t,0}$ and $q_{t,0}^{\mathrm{KDM}}$ at an anchor θ_0 while differentiating the model numerator and LEDH map. Younis and Sudderth do not derive this conditional LEDH construction. If an implementation instead recomputes the proposal as θ changes, it needs a separate derivation that couples the sampling map, proposal density, ancestor selection, and state map. Merely subtracting proposal or selection log-density tangents from a fixed-sample importance weight is wrong for the anchored model-normalizer derivative. This arm is therefore labelled `MODEL-IS` and is compared separately with both the canonical `ATOM-FINITE` score and the exact Kalman score.

At the atom reset itself, an ordinary density ratio cannot repair the change: a finite atom measure and an absolutely continuous KDM are mutually singular. A density-corrected KDM proposal can instead sample a continuous predictive update, where the physical transition supplies the target density. Inserting it directly at an atomic reset requires a different measure and an explicit new target; the predictive-density argument does not correct that reset.

7 Proposal-quality branch: controlled SMC, iAPF, covariance, and SGQF

The proposal question is separate from the score-identity question. A proposal can place particles in regions with large observation likelihood and thereby reduce weight variation, but that improvement is meaningful only if the physical target is left unchanged. The common one-step notation makes the boundary explicit. Given the incoming weighted cloud, let

$$u_t(x) = g_\theta(y_t \mid x) \sum_a W_{t-1}^a f_\theta(x \mid x_{t-1}^a) \tag{82}$$

be the unnormalised predictive target and let q_t be the actual marginal proposal density. Every proposal in this section uses

$$v_t^i = \frac{u_t(x_t^i)}{q_t(x_t^i)}, \qquad \widehat{z}_t = N^{-1} \sum_i v_t^i. \tag{83}$$

Changing q_t is allowed; replacing u_t by a Gaussian or quadrature approximation is a different filtering target.

7.1 Twisted particle filters and auxiliary lookahead

Let $\psi_t(x) > 0$ be a lookahead function. A twisted or auxiliary filter uses it to prefer ancestors and states that are expected to explain future observations. For example, define

$$\bar{\psi}_{t-1}(a) = \int f_\theta(x \mid x_{t-1}^a)\psi_t(x)\,\mathrm{d}x, \qquad r_t(a) \propto W_{t-1}^a \bar{\psi}_{t-1}(a), \tag{84}$$

and draw $A_t \sim r_t$ followed by $X_t \sim k_t(\cdot \mid A_t)$. The proposal is the joint law $r_t(a)k_t(x \mid a)$, or its marginal in x when the ancestor is integrated out. The importance ratio must therefore use the same joint or marginal denominator:

$$v_t(a, x) = \frac{W_{t-1}^a f_\theta(x \mid x_{t-1}^a) g_\theta(y_t \mid x)}{r_t(a)k_t(x \mid a)}. \tag{85}$$

Equivalent twisted-kernel formulations multiply a transition by a positive function and divide by its corresponding normalising factor [Whiteley and Lee, 2014]. These correction factors are what preserve the original likelihood. The ideal lookahead is a future-likelihood or eigenfunction object; a practical ψ_t is only an approximation to it. Consequently, twisting is a proposal-quality experiment, not a new score identity. For score diagnostics we freeze the fitted ψ_t and differentiate the declared finite target, while separately recording any derivative of the proposal that would be needed by a total-derivative claim.

The twisted branch is admissible for a degenerate transition only when the twist is defined on the same support and the required conditional law is available. A positive function on an ambient Euclidean space does not create a density for a transition supported on a lower-dimensional manifold.

7.2 Controlled SMC and the replicated lookahead tool

Controlled SMC provides a concrete way to learn the positive functions used in a twisted proposal. Heng et al. [2020, Sections 4.2–4.3, equations (20)–(21), Algorithms 2–3] approximate a backward dynamic-programming recursion by least-squares regression. The fitted quantity is the negative logarithm of a policy correction. Starting at the final observation and working backward, each regression includes the next fitted correction through an expectation under the current twisted transition. A subsequent SMC pass uses the refined policy, with the corresponding normalizing factors and potential corrections. Fitting a useful lookahead function and retaining these exact corrections are separate requirements: the former controls efficiency; the latter preserves the specified target.

This is the controlled-SMC method that has been replicated in the project. The October source checks establish agreement only within their declared released-source and input/failure scope; they are not a reproduction of all paper experiments. In particular, the recorded Lorenz covariance failure remains a failure, not a completed successful Lorenz experiment. Controlled SMC and the original iAPF both exploit twisting, but their fitted quantities and policy updates are not interchangeable. The controlled-SMC closure does not close the original-iAPF task discussed next.

For LEDH, the immediate proposal experiment is to use this learned lookahead to allocate ancestors or construct an evaluable proposal, while retaining the physical model in the numerator of (83) or (85). The actual joint or marginal sampling law determines the denominator. This integration still needs implementation and comparison; copying a controlled-SMC policy onto LEDH without its normalizers would not supply the stated correction. The fitted policy is frozen before a score comparison. Derivatives of the resulting parameter-dependent proposal must still be included wherever the declared finite-program derivative requires them. KDM can then provide an evaluable mixture or a mixture-expectation derivative within that specified calculation; its local importance-weight identity alone does not establish an unbiased normalized model score.

7.3 The iterated auxiliary particle filter

The iterated auxiliary particle filter (iAPF) constructs the lookahead rather than assuming that it is known [Guarniero et al., 2017]. A pilot particle run produces an approximation to the future value

function; a new auxiliary filter uses that approximation as ψ_t; and the procedure is iterated offline. Each iteration is still an auxiliary particle filter with an explicit proposal and correction of the form (85). The iteration may improve ancestor selection, but it does not make the pilot likelihood an exact normalizer and it does not provide an unbiased derivative of the marginal likelihood. The pilot data, number of iterations, and fitted ψ_t must be fixed before a claim run, or the adaptation must be included in a separate derivation.

In the LEDH implementation an iAPF can replace the UKF only in the proposal construction: it may supply r_t, proposal centres, or a lookahead covariance. The physical transition and observation factors remain in the numerator, and the complete proposal density (including ancestor probabilities and any LEDH Jacobian) remains in the denominator. The comparison therefore asks whether better lookahead improves coverage at equal cost; it does not ask whether iAPF is a replacement filtering theory for the target model.

The numerical fitting problem also needs a precise specification. In Section 5.1, equation (15), Guarniero et al. [2017] fit a Gaussian density to positive backward targets y_i by minimizing

$$L(m, \Sigma, \lambda) = \sum_{i=1}^{n} \{p_i(m, \Sigma) - \lambda y_i\}^2, \qquad p_i(m, \Sigma) = \mathcal{N}(x_i; m, \Sigma). \tag{86}$$

Here $y_i = g(x_i, y_t) f(x_i, \psi_{t+1})$ in their backward approximation; the symbol y_i in this fitting equation denotes a positive function value, not an observation. The scale λ accommodates an arbitrary positive normalization of that function. A log-density regression is a different criterion, even if it produces a useful Gaussian lookahead.

Proposition 3 (Profiled scale and possible absence of a minimizer)**.** *For fixed positive targets and a fixed nonsingular Gaussian, write $a = \sum_i p_i^2$, $b = \sum_i p_i y_i$, and $c = \sum_i y_i^2$. Then the unique optimal scale is $\lambda = b/c > 0$, and*

$$L = a - \frac{b^2}{c} + c\left(\lambda - \frac{b}{c}\right)^2. \tag{87}$$

Over unrestricted means and positive-definite covariances, the infimum of (86) *is zero, but it need not be attained.*

Proof. Expanding the squares gives $a - 2\lambda b + \lambda^2 c$; completing the square gives (87). Positivity of the targets and Gaussian densities gives $b, c > 0$. For the second claim, fix m and take $\Sigma = s^2 I_d$. Each density is bounded by $C_s = (2\pi)^{-d/2} s^{-d}$. Choosing $\lambda = C_s$ gives

$$0 \le L(m, s^2 I_d, C_s) \le C_s^2 \sum_i (1 + y_i)^2 \longrightarrow 0.$$

Since the loss is nonnegative, its infimum is zero. To see why a minimizer may be absent, take $d = 1$, $x_i \in \{-1, 0, 1\}$, and all three targets equal to one. A zero loss would require equal Gaussian densities at all three points. Equality at -1 and 1 forces $m = 0$; for every finite positive variance the density at zero is then strictly larger. Thus no finite Gaussian and scale attain zero in this example. □

The proposition identifies an ambiguity in an unrestricted global-argmin interpretation; it does not contradict the paper's reported numerical results. A local optimizer can return a useful finite fit. A compact mean region and diagonal standard deviations bounded above and away from zero also ensure that the profiled loss attains a minimum, by continuity on a compact set. Finding that minimum numerically remains a separate question. The implementation therefore states its bounds,

initialization, local solver and stopping rule. It records the covariance scale and the normalized shape residual $\sum_i (p_i - \widehat{\lambda} y_i)^2 / \sum_i p_i^2$ as well as the absolute loss: broadening a Gaussian can reduce its density amplitude without supplying an accurate lookahead.

For diagonal standard deviations σ_j, let $r_i = p_i - \widehat{\lambda} y_i$. The normal equation $\sum_i r_i y_i = 0$ cancels the derivative of the profiled scale. Differentiating the average loss $\overline{L} = L/n$ therefore gives

$$\frac{\partial \overline{L}}{\partial m_j} = \frac{2}{n} \sum_i r_i p_i \frac{x_{ij} - m_j}{\sigma_j^2}, \tag{88}$$

$$\frac{\partial \overline{L}}{\partial \log \sigma_j} = \frac{2}{n} \sum_i r_i p_i \left\{ \frac{(x_{ij} - m_j)^2}{\sigma_j^2} - 1 \right\}. \tag{89}$$

These are gradients of the density-scale criterion. Dividing all densities by a fixed positive constant merely rescales the objective; replacing it by the normalized shape residual as the optimization target would change the criterion. The shape residual is a diagnostic here.

Small density amplitudes also require care in floating-point arithmetic. Set $M = \max_i \log p_i$, $q_i = \exp(\log p_i - M)$ and $a = \sum_i q_i y_i / \sum_i y_i^2$. Then

$$\widehat{\lambda} = e^M a, \qquad L(m, \Sigma, \widehat{\lambda}) = e^{2M} \sum_i (q_i - a y_i)^2.$$

This evaluates the same density criterion. Computing $\log \widehat{\lambda}$ with log-sum-exp and the relative shape error from q_i avoids dividing two underflowed sums. The loss and its analytical gradient still retain the factor e^{2M}. If that factor falls below the numerical range, the implementation reports the underflow together with the fitted scale, active bounds and shape error. A tiny absolute loss or gradient can therefore end the numerical search without demonstrating an accurate lookahead.

The particle-count adaptation is a further part of the algorithm, rather than an implication of this fit. Algorithm 4 of Guarniero et al. [2017] starts with constant twists, checks the coefficient of variation of the latest $k+1$ likelihood estimates when the iteration index exceeds k, and may double the particle count when a complete equal-count window fails to increase monotonically. It then runs an independent final filter. Our bounded implementation declares the standard-deviation convention, finite iteration and particle caps, and behavior at an active fit bound. Exhausting a cap is a reported failure to converge. For the final finite-program score, the fitted coefficients and realized particle count are frozen; the score does not differentiate the offline stopping decision. These choices preserve the distinction between a fitted proposal, a derivative of its final random likelihood program, and the exact model score.

7.4 An exactly corrected nonlinear Gaussian twist

A nonlinear transition mean does not, by itself, prevent exact twisting. The relevant integral conditions on the previous state: its nonlinearity then appears in the location of a Gaussian density. This observation gives a tractable nonlinear comparator for the proposal study. It also separates the question of constructing a better proposal from the question of obtaining an unbiased model score.

Consider the scalar model

$$X_t = u_\theta(X_{t-1}) + \sqrt{Q_\theta}\, E_t, \qquad u_\theta(x) = a_\theta x + c \sin x, \tag{90}$$

$$Y_t = v_\theta(X_t) + \sqrt{R_\theta}\, V_t, \qquad v_\theta(x) = h_\theta x + b x^2, \tag{91}$$

where the noises are independent standard Gaussians, the initial law is μ_θ, and $Q_\theta, R_\theta > 0$. The curvature coefficients c, b are fixed in this comparison. Write $\varphi_s(z)$ for the centred Gaussian density

with variance s. Thus $f_t(z \mid x) = \varphi_{Q_\theta}(z - u_\theta(x))$ and $g_t(z) = \varphi_{R_\theta}(y_t - v_\theta(z))$, with the observations held fixed. These are the physical transition and observation densities, including their nonlinear means.

At each date choose a fitted lookahead

$$\psi_t(z) = \varphi_{B_t}(z - m_t) + \epsilon_t, \qquad B_t > 0, \quad \epsilon_t > 0. \tag{92}$$

The positive floor retains support wherever the physical transition has support. The lookahead need not integrate to one; it reweights a transition that will be normalized explicitly. Its coefficients may be fitted to each new observation sequence by the previously selected fitting procedure. The offline fitting draws are independent of the final filtering draws. Once fitting ends, its coefficients and the realized particle count are held fixed throughout the final value and score calculation. Oracle scores and final filtering errors play no role in fitting these coefficients.

Proposition 4 (Exact nonlinear conditional twist)**.** *For the model and lookahead above, set* $u = u_\theta(x)$, $Q = Q_\theta$, $S = Q + B_t$ *and*

$$M_t(x) = \int f_t(z \mid x)\psi_t(z)\,\mathrm{d}z = \varphi_S(m_t - u) + \epsilon_t. \tag{93}$$

Then $f_t^\psi(z \mid x) = f_t(z \mid x)\psi_t(z)/M_t(x)$ *is the mixture*

$$f_t^\psi(z \mid x) = (1 - \alpha_t(x))\varphi_Q(z - u) + \alpha_t(x)\varphi_{D_t}(z - \bar{m}_t), \tag{94}$$

where

$$\alpha_t(x) = \frac{\varphi_S(m_t - u)}{M_t(x)}, \qquad \bar{m}_t = u + \frac{Q}{S}(m_t - u), \qquad D_t = \frac{QB_t}{S}.$$

These expressions use the nonlinear $u_\theta(x)$ *exactly.*

Proof. Completing the square gives

$$\frac{(z-u)^2}{Q} + \frac{(z-m_t)^2}{B_t} = \frac{(z-\bar{m}_t)^2}{D_t} + \frac{(m_t-u)^2}{Q+B_t}.$$

The normalizing constants agree because $D_t(Q + B_t) = QB_t$. Consequently, the product of the two Gaussian densities is $\varphi_{Q+B_t}(m_t - u)\varphi_{D_t}(z - \bar{m}_t)$. Adding $\epsilon_t\varphi_Q(z - u)$ and integrating proves (93); dividing by that integral proves (94). □

Sampling this transition requires one component draw and one Gaussian draw. The first component explores the original transition, while the second moves toward the fitted lookahead centre. No derivative or linear approximation of u_θ enters this value calculation. When the floor dominates the Gaussian term, $\alpha_t(x)$ approaches zero and the sampler approaches the physical transition. Conversely, a sharply concentrated but inaccurate lookahead can place most samples poorly; tractable normalization does not guarantee quality.

The correction must also account for the lookahead inherited from the preceding date. Define $M_{T+1} = 1$ and

$$G_t^\psi(x) = \frac{g_t(x)M_{t+1}(x)}{\psi_t(x)}, \qquad C_0 = \int M_1(x)\,\mu_\theta(\mathrm{d}x), \qquad \mu_\theta^\psi(\mathrm{d}x) = \frac{M_1(x)}{C_0}\mu_\theta(\mathrm{d}x). \tag{95}$$

The factor M_{t+1} anticipates the next twisted transition. At the last date there is no such transition, which explains the terminal value one.

Proposition 5 (Preservation of the likelihood)**.** *The corrected twisted path law satisfies*

$$C_0\,\mu_\theta^\psi(\mathrm{d}x_0)\prod_{t=1}^{T}\{f_t^\psi(x_t\mid x_{t-1})G_t^\psi(x_t)\,\mathrm{d}x_t\}=\mu_\theta(\mathrm{d}x_0)\prod_{t=1}^{T}\{f_t(x_t\mid x_{t-1})g_t(x_t)\,\mathrm{d}x_t\}. \tag{96}$$

Its integral is therefore the original observed-data likelihood.

Proof. Substitute (95) and $f_t^\psi = f_t\psi_t/M_t$. At every date the two $\psi_t(x_t)$ factors cancel. The remaining multiplier of the physical path density is

$$M_1(x_0)\prod_{t=1}^{T}\frac{M_{t+1}(x_t)}{M_t(x_{t-1})}=M_{T+1}(x_T)=1.$$

The initial C_0 cancels the denominator in μ_θ^ψ. □

For a nonlinear u_θ, C_0 need not itself have a closed form, even when μ_θ is Gaussian. The particle implementation samples $X_0^i\sim\mu_\theta$, computes $\widehat{C}_0=N^{-1}\sum_i M_1(X_0^i)$, and resamples this initial cloud with probabilities proportional to $M_1(X_0^i)$. At date t it samples the mixture (94), records the average of $G_t^\psi(X_t^i)$, and resamples with those weights. Its likelihood estimate is

$$\widehat{Z}_N=\widehat{C}_0\prod_{t=1}^{T}\left\{N^{-1}\sum_{i=1}^{N}G_t^\psi(X_t^i)\right\}. \tag{97}$$

The exact path identity explains every correction in this estimator; it does not equate a finite initial cloud with μ_θ^ψ. Setting every ψ_t to one recovers the physical bootstrap particle filter. This is a useful independent check on both the value and its derivative.

The score calculation differentiates (97) with the fit held fixed. Let a dot denote a directional parameter derivative, including the accumulated state derivative $\dot{x}$. The nonlinear means contribute

$$\dot{u}=\dot{a}\,x+(a+c\cos x)\dot{x},\qquad \dot{v}=\dot{h}\,x+(h+2bx)\dot{x}. \tag{98}$$

In particular, freezing the fitted lookahead does not freeze the incoming particles or the physical model. At fixed m_t, B_t, ϵ_t, direct differentiation of the normalizer gives

$$\log\dot{M}_t(x)=\alpha_t(x)\left[\frac{m_t-u}{S}\dot{u}+\frac{1}{2}\left\{\frac{(m_t-u)^2}{S^2}-\frac{1}{S}\right\}\dot{Q}\right]. \tag{99}$$

Both the propagated-state derivative and the transition-variance derivative are necessary. For the observation and lookahead terms, put $r=y_t-v_\theta(x)$ and $\beta_t(x)=\varphi_{B_t}(x-m_t)/\psi_t(x)$. Then

$$\log\dot{g}_t(x)=\frac{r}{R}\dot{v}+\frac{1}{2}\left(\frac{r^2}{R^2}-\frac{1}{R}\right)\dot{R}, \tag{100}$$

$$\log\dot{\psi}_t(x)=\beta_t(x)\frac{m_t-x}{B_t}\dot{x}, \tag{101}$$

$$\log\dot{G}_t^\psi(x)=\log\dot{g}_t(x)+\log\dot{M}_{t+1}(x)-\log\dot{\psi}_t(x). \tag{102}$$

The last-date normalizer derivative is zero.

The Gaussian draw itself must be differentiated too. For the product component,

$$\dot{\tilde{m}}_t=\frac{B_t}{S}\dot{u}+\frac{B_t(m_t-u)}{S^2}\dot{Q},\qquad \dot{D}_t=\frac{B_t^2}{S^2}\dot{Q}, \tag{103}$$

so a fixed standard Gaussian draw e gives $\dot{X}_t=\dot{\tilde{m}}_t+\dot{D}_t e/(2\sqrt{D_t})$. For the physical component the corresponding derivative is $\dot{u}+\dot{Q}e/(2\sqrt{Q})$.

Proposition 6 (Derivative of the frozen-fit finite program)**.** *Suppose the physical parameters and initial sampling map are differentiable, all variances are positive, and no random uniform lies exactly on an ancestor or mixture selection boundary at the evaluation point. Hold fixed the lookahead coefficients, realized particle count, and random numbers. Propagate state derivatives through the selected ancestors and Gaussian components as above. The resulting derivative is*

$$\dot{\log \widehat{Z}}_N = \sum_i \omega_0^i \dot{\log M_1}(X_0^i) + \sum_{t=1}^{T} \sum_i \omega_t^i \dot{\log G_t^{\psi}}(X_t^i), \tag{104}$$

where ω_0^i and ω_t^i are the normalized initial and date-t weights before the corresponding resampling.

Proof. Away from a selection boundary the discrete labels are locally constant. The state tangent follows by the ordinary chain rule for the selected sampling map; a selected ancestor carries its existing tangent forward. For positive w_i, differentiation gives $\dot{\log}(N^{-1}\sum_i w_i) = \sum_i (w_i/\sum_j w_j)\dot{\log} w_i$. Apply this identity to each factor in (97) and use (98)–(103). □

This is the derivative of the frozen-fit finite program on its current selection branch. It omits the derivative of the offline fit and its stopping decision by definition. Moving selection boundaries can matter when differentiating an expectation, even though a fixed run lies on such a boundary with probability zero. Moreover, the logarithm creates the ratio-bias issue already discussed. Thus an exact conditional sampler, an exactly corrected path law, and a verified analytical finite-program derivative can coexist with a biased estimate of the model score. Comparing that estimate with a refined nonlinear likelihood reference remains necessary. Using this iAPF as a standalone comparator also does not establish that its fitted moments have been integrated into the LEDH covariance lifecycle; that is a separate proposal implementation and comparison.

7.5 Alternative covariance geometries

Covariance is a proposal geometry, not a correction to the model. The program will distinguish the following matrices at each proposal step:

$$P_t^Q = \text{the covariance supplied by the physical transition}, \tag{105}$$
$$P_t^{\mathrm{UKF}} = \text{the per-particle UKF covariance}, \tag{106}$$
$$P_t^{\mathrm{KDM}} = \text{the weighted covariance of the continuous mixture}, \tag{107}$$
$$P_t^{\mathrm{LEDH}} = \text{the local flow/GenUT covariance}, \tag{108}$$
$$P_t^{\mathrm{SGQF}} = \text{the covariance projected from a quadrature cloud}. \tag{109}$$

An optional hybrid has the form $P_t^{\mathrm{hyb}} = \alpha P_t^{\mathrm{UKF}} + (1-\alpha) P_t^{\mathrm{SGQF}}$, with α tuned on calibration data and frozen on the claim scope. Every matrix is checked for symmetry, positive definiteness, scale, and condition number. If it changes the sampling distribution, the resulting q_t and its derivatives must change consistently; a covariance label cannot be added to a weight after the samples have been drawn.

7.6 Structured Gaussian quadrature as a proposal source

The fixed structured Gaussian quadrature filter (SGQF) supplies another way to construct the moments used in a proposal. These moments must respond to each new observation. For clarity, consider additive, independent process and observation noise with covariances Q_θ and R_θ. At time

$t-1$, let a component have conditioned mean m_{t-1} and covariance $P_{t-1} = L_{t-1}L_{t-1}^{\mathsf{T}}$. A standardized quadrature rule $\{(\zeta_\ell, \omega_\ell)\}$, with $\sum_\ell \omega_\ell = 1$, gives nodes $x_{t-1,\ell} = m_{t-1} + L_{t-1}\zeta_\ell$. Propagating these nodes through a_θ, the deterministic part of the transition, gives $a_{t,\ell} = a_\theta(x_{t-1,\ell})$ and the prediction

$$m_t^- = \sum_\ell \omega_\ell a_{t,\ell}, \qquad P_t^- = Q_\theta + \sum_\ell \omega_\ell (a_{t,\ell} - m_t^-)(a_{t,\ell} - m_t^-)^{\mathsf{T}}. \tag{110}$$

For nonadditive noise, the quadrature must instead cover the joint state and noise law; simply adding Q_θ would then compute the wrong prediction.

Conditioning is a second operation. Form fresh quadrature nodes $x_{t,\ell}^- = m_t^- + L_t^-\zeta_\ell$, where $L_t^-(L_t^-)^{\mathsf{T}} = P_t^-$, and evaluate $b_{t,\ell} = b_\theta(x_{t,\ell}^-)$, the observation mean map. Define

$$\begin{aligned}
\bar{y}_t &= \sum_\ell \omega_\ell b_{t,\ell}, \qquad S_t = R_\theta + \sum_\ell \omega_\ell (b_{t,\ell} - \bar{y}_t)(b_{t,\ell} - \bar{y}_t)^{\mathsf{T}}, \\
C_t &= \sum_\ell \omega_\ell (x_{t,\ell}^- - m_t^-)(b_{t,\ell} - \bar{y}_t)^{\mathsf{T}}, \qquad K_t = C_t S_t^{-1}, \\
m_t &= m_t^- + K_t(y_t - \bar{y}_t), \qquad P_t = P_t^- - K_t S_t K_t^{\mathsf{T}}.
\end{aligned} \tag{111}$$

This is an assumed-Gaussian conditioning approximation, exact for the linear-Gaussian model when the rule integrates the required second moments. Likelihood-reweighting the nodes is a different approximation and must be identified as such. In particular, a sparse quadrature rule may have negative weights. These are integration weights, not categorical sampling probabilities; normalising them does not make them valid resampling weights. Positive definiteness of both S_t and the proposed P_t must be checked.

The conditioned component state (m_t, P_t) is carried into the next prediction through the declared component-assignment and reset operation. When an OT reset changes that assignment, the covariance provider must consume the actual reset state and propagate its derivatives. Reusing a fixed (m_0, P_0) at every date would omit observation dependence. The analytical score therefore needs the derivatives of the initial moments, propagated nodes, noise covariances, conditioning solve, and subsequent reset, including their dependence on θ. This lifecycle applies separately to every component if a mixture supplies the proposal.

These moments can parameterise a Gaussian, mixture, or LEDH proposal. The repository interfaces `tf_fixed_sgqf_cloud` and `tf_fixed_sgqf_filter` provide the cloud and filtering moments; the current route is an eager/Python-branch, approximate proposal-design lane. Its sparse-grid point count and memory must be recorded as dimension and level increase. It is not a claim-bearing replacement for the physical filter, and its approximate likelihood must never be used in place of u_t in (83).

The first SGQF experiment therefore takes the conditioned m_t and P_t to define q_t^{SGQF}, evaluates the exact model numerator and proposal denominator, and compares weight tails with UKF, KDM, and LEDH moments. A high-dimensional result is admissible only after the call chain, batching, device, and scaling checks show that the proposal is the one actually used by the claim-bearing lane.

7.7 Systematic proposal test

The proposal study is a factorial comparison, with the score estimator held fixed while proposal quality is measured. Its proposal families are bootstrap, locally adapted, KDM, LEDH/GenUT, twisted PF, iAPF, SGQF-guided, and defensive mixtures. Crossed factors are lookahead (none, fixed one-step, iterated ψ), covariance (P^Q, UKF, KDM, LEDH, SGQF, and calibrated hybrid), particle count, horizon, and observation regime. Model-corrected normalizer MSE is the primary

criterion for a normalizer-accuracy claim. A better-score claim requires lower oracle score MSE at the declared particle or compute budget, after the proposal identity and target-preservation tests pass. Coefficient of variation of $\widehat{z}_t$, effective sample size, and weight tails explain the result; they do not replace score error as the criterion. The linear-Gaussian/Kalman model is the first oracle-bearing check; nonlinear regular-transition models and high-dimensional SGQF cases follow.

Hard vetoes are a missing or non-integrable proposal density, support failure, non-finite or incorrectly corrected weights, use of an approximate SGQF or twisted likelihood as the target, and a claim route that cannot be traced to the declared proposal. A paired uncertainty analysis is required before calling a candidate better. A passing proposal test establishes a viable proposal arm; it does not establish an unbiased score.

8 Directional finite-difference validation ladder

Finite differences provide an independent directional check on a score, not an automatic bias oracle. For a unit direction v, define $s_v = v^{\mathsf{T}}\nabla_\theta \ell(\theta)$ and evaluate the same particle likelihood program at a fixed symmetric set of perturbations. The three-point central estimate is

$$D_2(h;v) = \frac{\ell(\theta+hv) - \ell(\theta-hv)}{2h}, \qquad D_2(h;v) = s_v + O(h^2), \tag{112}$$

and the five-point fourth-order estimate is

$$D_4(h;v) = \frac{\ell(\theta-2hv) - 8\ell(\theta-hv) + 8\ell(\theta+hv) - \ell(\theta+2hv)}{12h}, \qquad D_4(h;v) = s_v + O(h^4). \tag{113}$$

For an eleven-point cubic fit, let $k = -5, \ldots, 5$ and fit

$$(\widehat{\beta}_0, \ldots, \widehat{\beta}_3) = \arg\min_{\beta} \sum_{k=-5}^{5} \left[\widehat{\ell}(\theta + khv) - \sum_{r=0}^{3} \beta_r (kh)^r \right]^2, \qquad D_3(h;v) = \widehat{\beta}_1. \tag{114}$$

Write $f(u) = \ell(\theta + uv)$. A bounded third derivative near zero suffices for the second-order statement in (112). A bounded fifth derivative, with $f \in C^5$, suffices for the fourth-order statement in (113) and for the symmetric cubic fit. Four continuous derivatives alone do not suffice: $f(u) = \operatorname{sign}(u)|u|^{4+\alpha}$, $0 < \alpha < 1$, has four continuous derivatives but gives derivative error of order $h^{3+\alpha}$ for the five-point rule.

Proposition 7 (Symmetric cubic derivative weights)**.** *For the eleven equally spaced nodes in* (114)*, define* $S_r = \sum_{j=-5}^{5} j^r$*. The fitted derivative is*

$$D_3(h;v) = \frac{1}{h} \sum_{j=-5}^{5} c_j \widehat{\ell}(\theta + jhv), \qquad c_j = \frac{S_6 j - S_4 j^3}{S_2 S_6 - S_4^2}, \qquad (S_2, S_4, S_6) = (110, 1958, 41030). \tag{115}$$

For a deterministic $f \in C^5$*, the leading errors are* $D_4 - f'(0) = -f^{(5)}(0)h^4/30 + o(h^4)$ *and* $D_3 - f'(0) = -143 f^{(5)}(0)h^4/90 + o(h^4)$*.*

Proof. Symmetry separates the odd and even columns of the least-squares design. Fitting against the dimensionless columns j and j^3 gives the odd normal matrix $\left(\begin{smallmatrix} S_2 & S_4 \\ S_4 & S_6 \end{smallmatrix}\right)$. Its first inverse row gives c_j; division by h converts the fitted coefficient to a derivative. Hence $\sum c_j = 0$, $\sum j c_j = 1$, $\sum j^2 c_j = \sum j^3 c_j = \sum j^4 c_j = 0$. Taylor expansion leaves $h^4 f^{(5)}(0) \sum j^5 c_j / 5!$. The last sum divided by 5! is $-143/90$. Applying the same calculation to the coefficients $(1, -8, 0, 8, -1)/12$ at $j = (-2, -1, 0, 1, 2)$ gives $-1/30$. □

Thus more points do not give a higher formal order. The eleven-point cubic fit has a different truncation constant and a different response to correlated likelihood noise; a symmetric linear or quadratic fit generally retains an $O(h^2)$ term.

Proposition 8 (Finite-difference mean squared error)**.** *Let $\widehat{L}_j$ be the random log-likelihood estimate at node j, with finite second moments and joint covariance matrix Σ. For coefficients c that reproduce a first derivative, put*

$$T_h = \frac{1}{h}\sum_j c_j \ell(\theta + jhv) - s_v, \qquad B_{N,h} = \frac{1}{h}\sum_j c_j \{\mathbb{E}\widehat{L}_j - \ell(\theta + jhv)\}. \tag{116}$$

Then

$$\mathbb{E}\left[\left(\frac{1}{h}\sum_j c_j \widehat{L}_j - s_v\right)^2\right] = (T_h + B_{N,h})^2 + \frac{c^\mathsf{T}\Sigma c}{h^2}. \tag{117}$$

Proof. The estimator's mean error is $T_h + B_{N,h}$. Its centered part is $h^{-1}c^\mathsf{T}(\widehat{L} - \mathbb{E}\widehat{L})$, with variance $h^{-2}c^\mathsf{T}\Sigma c$. Expanding the squared error, the mixed term with the centered part has zero expectation. The squared bias still contains $2T_h B_{N,h}$; the truncation and particle biases need not have the same sign. □

A universal U-shaped error curve does not follow. Shared additive noise cancels because $\sum_j c_j = 0$. A mean-square differentiable coupled random likelihood can have a bounded derivative variance as $h \downarrow 0$. Independent node noise with nonvanishing variance instead yields the familiar h^{-2} variance growth. The joint coupling, as well as the stencil, matters.

Use a deterministic positive ladder $\mathcal{H} = \{h_0, h_0/2, \ldots\}$, with h_0 in parameter units and v normalised. An unrestricted normal draw of h is excluded because values close to zero divide likelihood noise by a small number. A truncated positive distribution is a later randomized arm, not the default; the negative evaluations are the paired points in the same symmetric stencil. Common random numbers or a declared coupling are used to reduce variance, but they do not remove finite-particle, log-normalisation, or truncation bias.

For a d-parameter score, use d independent coordinate directions or a full-row-rank matrix $V = (v_1, \ldots, v_m) \in \mathbb{R}^{d\times m}$. With directional estimates b, the least-squares solution is

$$\widehat{s} = (VV^\mathsf{T})^{-1}Vb. \tag{118}$$

Implementation uses QR or SVD rather than an explicit inverse. Singular values, directional-error propagation, parameter scale and dtype determine whether reconstruction is informative; an arbitrary condition-number cutoff does not establish that. Every perturbed parameter must remain in the declared model domain and be distinct at the working precision. If directions are defined in transformed coordinates, the reported likelihood score must include the corresponding coordinate chain rule.

The experiment has two separate checks. Deterministic analytic functions or exact Kalman log likelihoods test truncation slopes near two or four before roundoff dominates. Stochastic particle runs estimate the full error and replicate covariance in (117); those curves need not have either slope. Direct averages of squared oracle errors estimate MSE, whereas squaring a noisy sample mean does not give an unbiased squared-bias estimate. Calibrate the positive ladder and coupling on oracle-bearing data, validate and freeze them, then evaluate on untouched data. Without an oracle the ladder remains a diagnostic or parity calculation; agreement alone cannot certify model-score bias or supply a known control-variate center.

9 A model score that averages over ancestors

A well-chosen proposal can improve the particle approximation, but a separate question remains: which score should we compute from those particles? Differentiating every operation of a chosen finite filter is one answer. Fisher's identity supplies another, aimed directly at the observed-data score. The recursion below is a smoothing construction, even though its history averages are updated forward in time. It requires evaluable transition densities and a valid backward conditional. It does not apply as written to the degenerate DSGE transitions discussed in Section 9.1.

Assume the initial, transition, and observation laws have differentiable densities on a common dominating measure, and that differentiation may pass through the joint integral. For $x_{0:T}$, the joint density is

$$p_\theta(x_{0:T}, y_{1:T}) = \mu_\theta(x_0) \prod_{t=1}^{T} f_\theta(x_t \mid x_{t-1}) g_\theta(y_t \mid x_t). \tag{119}$$

The derivative inside this density holds $x_{0:T}$ fixed. Define its complete data score by

$$H_T(x_{0:T}) = \nabla_\theta \log \mu_\theta(x_0) + \sum_{t=1}^{T} \left\{ \nabla_\theta \log f_\theta(x_t \mid x_{t-1}) + \nabla_\theta \log g_\theta(y_t \mid x_t) \right\}. \tag{120}$$

Since $\nabla_\theta p_\theta = p_\theta H_T$, integrating this equality and dividing by $Z = p_\theta(y_{1:T})$ gives

$$\begin{aligned} s(\theta) &= \frac{\int p_\theta(x_{0:T}, y_{1:T}) H_T(x_{0:T}) \, \mathrm{d}x_{0:T}}{Z} \\ &= \mathbb{E}_\theta[H_T(X_{0:T}) \mid y_{1:T}]. \end{aligned} \tag{121}$$

The posterior expectation is over whole state histories, including the initial state. A parameter-dependent stationary initial law contributes the first term even when no observation is assigned to time zero. In a latent-innovation representation, the density and the derivative must instead be derived in those coordinates; the state-density expression cannot be used unchanged for a singular transition.

The needed history average has a forward recursion. Let H_t be (120) truncated at time t, and define $A_t(x_t) = \mathbb{E}[H_t(X_{0:t}) \mid X_t = x_t, y_{1:t}]$. Conditional on X_t, the observation at t does not further distinguish its ancestors. The backward conditional distribution is proportional to $p_\theta(x_{t-1} \mid y_{1:t-1}) f_\theta(x_t \mid x_{t-1})$. Conditioning the additive score on this distribution gives

$$\begin{aligned} A_t(x) = {} & \nabla_\theta \log g_\theta(y_t \mid x) \\ & + \int \left[A_{t-1}(z) + \nabla_\theta \log f_\theta(x \mid z) \right] p_\theta(z \mid x, y_{1:t-1}) \, \mathrm{d}z. \end{aligned} \tag{122}$$

This is the step that a single realized ancestor cannot represent exactly. It averages all histories compatible with the current state instead of discarding them when a resampling label is selected.

For a weighted particle approximation, define the backward probabilities

$$b_t^{ji} = \frac{W_{t-1}^j f_\theta(x_t^i \mid x_{t-1}^j)}{\sum_k W_{t-1}^k f_\theta(x_t^i \mid x_{t-1}^k)}. \tag{123}$$

All terms are evaluated at the current parameter; the denominator must be positive. The initial particles must approximate μ_θ: they may be drawn from it directly, or drawn from a covering

proposal and weighted by the corresponding initial-law ratio. Initialize $A_0^j = \nabla_\theta \log \mu_\theta(x_0^j)$ and use

$$A_t^i = \nabla_\theta \log g_\theta(y_t \mid x_t^i) + \sum_j b_t^{ji} \left[A_{t-1}^j + \nabla_\theta \log f_\theta(x_t^i \mid x_{t-1}^j) \right], \tag{124}$$

$$\widehat{s}_N = \sum_i W_T^i A_T^i. \tag{125}$$

This is the marginal-score construction in Poyiadjis et al. [2011, Section 2.2, Algorithm 2, equations (20) and (22)], with the initial state indexed explicitly as time zero. Its relation to modified differentiation rules is discussed by Ścibior and Wood [2021, Section 4.2, equations (24)–(25)]. The displayed recursion follows directly from the conditional expectation above, with the initial and observation timing made explicit.

No derivative of the proposal or of the sampled trajectory is missing from (125): its integrand is the complete-data model score. Proposal design changes how well its expectation is estimated. This differs from S_0^N, where state motion through the executed finite program is part of the claimed derivative. The Fisher estimator should therefore be checked against a model-score oracle and checked local density derivatives. Requiring it to match finite differences of the old OT-reset scalar would ask it to compute a different quantity.

Particle approximation and normalization still introduce error. With an exact smoothing draw, H_T is unbiased for s under the stated integrability conditions; with a finite approximate smoothing distribution, $\widehat{s}_N$ is not generally unbiased. Averaging ancestors reduces a particular source of randomness without proving a universal ranking of full filters. This remains true when the marginal likelihood estimate itself is unbiased.

The original Poyiadjis paper gives a useful, specific precedent. Its Section 2.3 compares the methods against an analytic Gaussian-model score using 500 particles and 100 replications. The discussion of linear growth in the marginal method's variance combines a stability conjecture with those numerical results; Theorem 1 instead proves a quadratic lower bound for the genealogical construction under its stated mixing and boundedness conditions. These distinctions prevent a local ancestor-averaging argument from becoming an unsupported universal variance theorem for the proposed filter.

There is a useful computational overlap with Section 6. The transition mixture in the numerator of (71) is also the denominator of (123). The same pairwise densities can support both the value calculation and the score recursion. A direct implementation costs $O(TN^2)$, plus the cost of the model derivatives. Streaming reduces memory use, not the number of pairs. PaRIS is an option for replacing the exact ancestor sum by a controlled number of backward draws; its acceptance cost, added variance, and support assumptions must be measured in the actual model before choosing it over the exact reference.

9.1 Why degenerate transitions obstruct this recursion

Consider a state equation with fewer independent shocks than state coordinates,

$$x_t = A_\theta x_{t-1} + G_\theta \varepsilon_t, \qquad \varepsilon_t \sim N(0, I_r), \tag{126}$$

$$\mathcal{S}_j = A_\theta x_{t-1}^j + \operatorname{range}(G_\theta), \qquad \operatorname{rank}(G_\theta) = r < d_x. \tag{127}$$

Here $x_t \in \mathbb{R}^{d_x}$, and $\mathcal{S}_j$ is the set of successors allowed by ancestor j. Each conditional transition is supported on that affine subspace and has no full-dimensional Lebesgue density. Thus (123) cannot be evaluated with ordinary ambient Gaussian densities. A pseudoinverse and pseudodeterminant alone do not repair it: a rank-aware density must also enforce the correct support.

The AR(2) example in (174) makes the failure concrete. A child drawn from ancestor a satisfies $x_{t,2} = x^a_{t-1,1}$. At a fixed parameter, conditioning the finite mixture on this supporting line excludes every ancestor with a different first coordinate. If these coordinates are distinct, the compatible ancestor probabilities are

$$b_t^{ji} = \mathbf{1}\{j = a\}. \tag{128}$$

The proposed history average then collapses to the sampled genealogy. It does not recover the alternative histories needed to repair genealogical degeneracy. PaRIS changes how backward ancestors are sampled; it cannot give positive probability to these incompatible transitions.

This obstruction concerns the conditional transitions between fixed particles. The exact predictive marginal can still be full dimensional after integrating over earlier continuous state uncertainty, as the same AR(2) example shows. That fact does not make the finite mixture of distinct conditional supports full dimensional. An ambient Gaussian proposal assigns zero probability to their union, so it cannot provide the exact importance correction for that finite singular mixture. Adding full-dimensional process noise changes the model; smoothing the finite predictive mixture into an ambient KDM changes the approximation. Neither operation establishes the displayed model-corrected recursion for the original degenerate transition.

Innovation coordinates are a possible starting point for a different derivation. They require the joint likelihood to be written in those coordinates, including the initial law and the dependence of the state map on θ. Substituting an innovation density for $f_\theta(x_t \mid x_{t-1})$ in the existing backward weights is not such a derivation. In particular, moving support contributes state-map derivatives to the observation factors, as (178) demonstrates.

Some degenerate models admit an exact reduction to common stochastic coordinates, or another specially derived smoothing method. The obstruction therefore does not rule out every form of smoothing. It does rule out the ordinary all-ancestor recursion above as a general method for the degenerate DSGE target. A successful full-rank Gaussian comparison would establish no extension to that target. The disturbance construction below instead needs its own support and differentiation assumptions.

9.2 A disturbance-coordinate proposal and its score

The support obstruction changes the coordinates in which the proposal must be defined. Instead of proposing an ambient state and then asking for the density of a singular transition, propose the random variables that generate the state. This is the disturbance-state-space formulation of Murray et al. [2013, Sections 1–2]. It is also the only formulation used in the propositions below.

Let $v_0 \in \mathsf{V}_0$ be an initial disturbance and let $u_t \in \mathsf{U}_t$ be the innovation at time t. The sets V_0 and U_t carry fixed reference measures λ_0 and λ_t; they need not have the same dimension as the state. For a parameter θ in an open set $\Theta \subset \mathbb{R}^p$, define

$$x_0 = F_{0,\theta}(v_0), \qquad x_t = F_{t,\theta}(x_{t-1}, u_t), \quad t = 1, \dots, T, \tag{129}$$

$$y_t \mid x_t \sim g_{t,\theta}(\cdot \mid x_t), \qquad v_0 \sim \rho_{0,\theta}, \quad u_t \mid x_{t-1} \sim r_{t,\theta}(\cdot \mid x_{t-1}). \tag{130}$$

The transition may be deterministic and rank deficient in x_t; no density with respect to ambient Lebesgue measure on $\mathbb{R}^{d_x}$ is assumed. The densities $\rho_{0,\theta}$, $r_{t,\theta}$, and $g_{t,\theta}$ are with respect to their displayed fixed reference measures; the observation reference measure is also independent of θ. We assume that the innovations have the displayed conditional Markov law given the past and that observations are conditionally independent given the state path and θ. These are model assumptions. If shocks are independent of the state and parameter, $r_{t,\theta}$ reduces to a fixed r_t.

Assumption 1 (Disturbance regularity)**.** *For $\lambda = \lambda_0 \otimes \lambda_1 \otimes \cdots \otimes \lambda_T$, define $z = (v_0, u_{1:T})$ and let $x_{0:T}(z, \theta)$ be generated by* (129)*. There is a neighbourhood B of the parameter value under study such that:*

(i) on a common set of full λ measure the state maps are continuously differentiable in their parameter and state arguments along the generated paths, and the displayed density factors are positive and continuously differentiable there. Their common disturbance support is independent of $\theta \in B$;

(ii) the disturbance-coordinate integrand

$$\pi_\theta(z; y) = \rho_{0,\theta}(v_0) \prod_{t=1}^{T} r_{t,\theta}(u_t \mid x_{t-1}) g_{t,\theta}(y_t \mid x_t) \tag{131}$$

has a positive finite integral $Z(\theta; y)$ for $\theta \in B$; and

(iii) each component of $\nabla_\theta \pi_\theta(z; y)$ is dominated on B by an integrable envelope. The same conditions hold for each time prefix used below.

The fixed-coordinate condition is the important one. A state support that moves with θ is harmless when it is generated by the same fixed disturbance coordinate z; differentiating an ambient state density is not needed. For a parameter-dependent shock support, first introduce a fixed-base variable and put its parameter dependence into $F_{t,\theta}$ and its density.

Use column gradients for scalar functions and let $\dot{x}_t$ be the $d_x \times p$ Jacobian of x_t with respect to θ, holding the disturbances fixed. Define this total tangent recursively by

$$\dot{x}_0 = \partial_\theta F_{0,\theta}(v_0), \tag{132}$$

$$\dot{x}_t = \partial_\theta F_{t,\theta}(x_{t-1}, u_t) + J_x F_{t,\theta}(x_{t-1}, u_t) \dot{x}_{t-1}. \tag{133}$$

For any state-dependent shock density, use the total derivative

$$D_\theta \log r_{t,\theta}(u_t \mid x_{t-1}) = \partial_\theta \log r_{t,\theta}(u_t \mid x_{t-1}) + \dot{x}_{t-1}^{\mathsf{T}} \nabla_x \log r_{t,\theta}(u_t \mid x_{t-1}). \tag{134}$$

The complete disturbance-path score is

$$\begin{aligned} h_\theta(z; y) = \nabla_\theta \log \rho_{0,\theta}(v_0) &+ \sum_{t=1}^{T} D_\theta \log r_{t,\theta}(u_t \mid x_{t-1}) \\ &+ \sum_{t=1}^{T} \left[\partial_\theta \log g_{t,\theta}(y_t \mid x_t) + \dot{x}_t^{\mathsf{T}} \nabla_x \log g_{t,\theta}(y_t \mid x_t) \right]. \end{aligned} \tag{135}$$

Proposition 9 (Disturbance-coordinate Fisher identity)**.** *Under Assumption 1, the observed-data likelihood and score are*

$$Z(\theta; y) = \int \pi_\theta(z; y) \, \mathrm{d}\lambda(z), \tag{136}$$

$$\nabla_\theta \log Z(\theta; y) = \mathbb{E}_\theta[h_\theta(\mathsf{Z}; y) \mid y_{1:T}], \tag{137}$$

where the conditional law in the second line has density $\pi_\theta(z; y)/Z(\theta; y)$ with respect to λ.

Proof. The recursion in (129) makes $\pi_\theta(z; y)$ a function on the fixed measure space $(\mathsf{V}_0 \times \mathsf{U}_{1:T}, \lambda)$. By the envelope condition, differentiation may pass through the integral, so

$$\nabla_\theta Z(\theta; y) = \int \nabla_\theta \pi_\theta(z; y) \, \mathrm{d}\lambda(z). \tag{138}$$

At points where the factors are positive, the product rule gives $\nabla_\theta \pi_\theta = \pi_\theta h_\theta$. The chain rule through the state recursion gives (133); applying it to each shock and observation factor gives (135). Substitution into (138), followed by division by the positive $Z(\theta; y)$, gives

$$\nabla_\theta \log Z(\theta; y) = \int h_\theta(z; y) \frac{\pi_\theta(z; y)}{Z(\theta; y)} \, \mathrm{d}\lambda(z), \tag{139}$$

which is the asserted conditional expectation. No transition density in the ambient state coordinates appears in the argument. □

The initial term in (135) is not a cosmetic addition. If the model starts in its stationary distribution and that law depends on θ, the stationary derivative belongs in $\nabla_\theta \log \rho_{0,\theta}$. If instead a parameter-independent disturbance generates the initial state through $F_{0,\theta}$, its effect appears in $\dot{x}_0$ and in every later observation term.

Proposition 10 (Support-valid sequential proposal)**.** *Under Assumption 1, suppose an initial proposal $q_{0,\theta}(v_0)$ and, for each t, a proposal $q_{t,\theta}(u_t \mid x_{t-1}, y_t)$ are positive wherever the corresponding target factors in* (131) *are positive. For sequential importance sampling without resampling, draw from*

$$Q_\theta(z) = q_{0,\theta}(v_0) \prod_{t=1}^{T} q_{t,\theta}(u_t \mid x_{t-1}, y_t), \tag{140}$$

and set, for a draw $\mathsf{Z} \sim Q_\theta$,

$$R_\theta(z) = \frac{\pi_\theta(z; y)}{Q_\theta(z)}, \tag{141}$$

$$\widehat{Z} = R_\theta(\mathsf{Z}), \qquad \widehat{D} = R_\theta(\mathsf{Z}) h_\theta(\mathsf{Z}; y). \tag{142}$$

The differentiability and integrable derivative bound in Assumption 1 imply that Z is differentiable and that $\pi_\theta h_\theta$ is integrable. Consequently,

$$\mathbb{E}_{Q_\theta}[\widehat{Z}] = Z(\theta; y), \qquad \mathbb{E}_{Q_\theta}[\widehat{D}] = \nabla_\theta Z(\theta; y). \tag{143}$$

Where $\widehat{Z} > 0$, the one-draw ratio is $h_\theta(\mathsf{Z}; y)$. Its sampling law here is Q_θ, so it is generally not an unbiased model-score estimator. If Q_θ is the exact disturbance posterior, even one such draw is unbiased by Proposition 9; an arbitrary proposal has no such guarantee.

Proof. The support condition makes R_θ well defined Q_θ-almost everywhere. Cancellation of Q_θ gives

$$\mathbb{E}_{Q_\theta}[R_\theta] = \int \pi_\theta(z; y) \, \mathrm{d}\lambda(z) = Z(\theta; y),$$

$$\mathbb{E}_{Q_\theta}[R_\theta h_\theta] = \int \pi_\theta(z; y) h_\theta(z; y) \, \mathrm{d}\lambda(z) = \nabla_\theta Z(\theta; y),$$

where the last equality is Proposition 9 before division by Z. The proposal density may depend on θ: it cancels in the pointwise importance ratio. A derivative of Q_θ is needed only if one differentiates the proposal expectation itself, rather than using the score identity to estimate $\nabla_\theta Z$. □

For a particle filter, the same ratio is applied one step at a time. Let $\alpha_t(a \mid \mathcal{F}_{t-1})$ be an ancestor proposal, positive whenever $W_{t-1}^a > 0$, and draw u_t^i from $q_{t,\theta}$ after drawing a_t^i. The support-valid auxiliary particle weight is

$$\omega_t^i = \frac{W_{t-1}^{a_t^i}}{\alpha_t(a_t^i \mid \mathcal{F}_{t-1})} \frac{r_{t,\theta}(u_t^i \mid x_{t-1}^{a_t^i}) g_{t,\theta}(y_t \mid x_t^i)}{q_{t,\theta}(u_t^i \mid x_{t-1}^{a_t^i}, y_t)}, \qquad x_t^i = F_{t,\theta}(x_{t-1}^{a_t^i}, u_t^i). \tag{144}$$

If $\alpha_t = W_{t-1}$, the first ratio is one. A lookahead or guided ancestor law changes α_t but not the numerator. This is the exact place where a Younis or UKF construction may enter: it supplies α_t or q_t while the disturbance density and observation likelihood remain those of the declared model.

Proposition 11 (Conditional APF normalizer identity)**.** *Conditionally on $\mathcal{F}_{t-1}$, assume that α_t and $q_{t,\theta}$ are fixed before the ancestor and innovation draws and that the ratios in* (144) *are integrable. Then, for every particle,*

$$\mathbb{E}[\omega_t^i \mid \mathcal{F}_{t-1}] = \sum_{a=1}^N W_{t-1}^a \int r_{t,\theta}(u \mid x_{t-1}^a) g_{t,\theta}(y_t \mid F_{t,\theta}(x_{t-1}^a, u)) \, \mathrm{d}\lambda_t(u). \tag{145}$$

If the incoming empirical measure is replaced by the exact filtering measure, the right side is the exact predictive likelihood.

Proof. Condition on an ancestor value a. The proposal density cancels in the innovation integral:

$$\mathbb{E}[\omega_t^i \mid \mathcal{F}_{t-1}, a_t^i = a] = \frac{W_{t-1}^a}{\alpha_t(a \mid \mathcal{F}_{t-1})} \int r_{t,\theta}(u \mid x_{t-1}^a) g_{t,\theta}(y_t \mid F_{t,\theta}(x_{t-1}^a, u)) \, \mathrm{d}\lambda_t(u).$$

Taking expectation over $a_t^i \sim \alpha_t$ yields (145). Integrating the same innovation likelihood over the exact incoming filtering measure gives the predictive likelihood by the law of total probability. □

The one-step cancellation alone is not a proof about a complete filtering run. To supply that proof, retain the disturbance history $z_{0:t} = (v_0, u_{1:t})$ mathematically and write $\gamma_t(\varphi) = \int \pi_{\theta,t} \varphi \, \mathrm{d}\lambda_{0:t}$, where $\pi_{\theta,t}$ is (131) stopped at t. There is no observation at time zero, so $\pi_{\theta,0} = \rho_{0,\theta}$ and $\gamma_0(1) = 1$. The extension kernel for a history test function is

$$(K_{t,\theta}\varphi)(z_{0:t-1}) = \int r_{t,\theta}(u \mid x_{t-1}) g_{t,\theta}(y_t \mid F_{t,\theta}(x_{t-1}, u)) \varphi(z_{0:t-1}, u) \, \mathrm{d}\lambda_t(u). \tag{146}$$

Direct integration of the last disturbance gives $\gamma_t(\varphi) = \gamma_{t-1}(K_{t,\theta}\varphi)$.

Proposition 12 (Unbiased particle likelihood and derivative)**.** *Fix $N \geq 1$ and θ. Initially draw v_0^i independently from $q_{0,\theta}$ and set $\omega_0^i = \rho_{0,\theta}(v_0^i)/q_{0,\theta}(v_0^i)$. At each subsequent time, draw ancestors conditionally independently from α_t and innovations from the proposals in* (144)*. Suppose all weights are positive and finite almost surely, all proposal support conditions hold, and the test functions below are absolutely integrable under γ_t. Define*

$$\overline{\omega}_t = \frac{1}{N} \sum_i \omega_t^i, \qquad W_t^i = \frac{\omega_t^i}{N \overline{\omega}_t}, \qquad \widehat{Z}_t = \prod_{s=0}^t \overline{\omega}_s, \qquad \gamma_t^N(\varphi) = \widehat{Z}_t \sum_i W_t^i \varphi(z_{0:t}^i). \tag{147}$$

For every deterministic test function φ with the stated integrability, including one that depends on the fixed θ,

$$\mathbb{E}[\gamma_t^N(\varphi)] = \gamma_t(\varphi). \tag{148}$$

In particular, with $h_{\theta,t} = \nabla_\theta \log \pi_{\theta,t}$,

$$\widehat{D}_t = \widehat{Z}_t \sum_i W_t^i h_{\theta,t}(z_{0:t}^i), \qquad \mathbb{E}\widehat{Z}_t = Z_t, \qquad \mathbb{E}\widehat{D}_t = \nabla_\theta Z_t. \tag{149}$$

Proof. At time zero, cancellation of the initial proposal gives

$$\mathbb{E}\gamma_0^N(\varphi) = \frac{1}{N}\sum_i \mathbb{E}\left[\frac{\rho_{0,\theta}(v_0^i)}{q_{0,\theta}(v_0^i)}\varphi(v_0^i)\right] = \int \rho_{0,\theta}\varphi \,\mathrm{d}\lambda_0 = \gamma_0(\varphi).$$

For the induction step, the definitions cancel the random normalization:

$$\gamma_t^N(\varphi) = \widehat{Z}_{t-1}\frac{1}{N}\sum_i \omega_t^i \varphi(z_{0:t}^i).$$

The ancestor and proposal cancellations used in Proposition 11, now with φ in the integrand, give the exact conditional identity

$$\mathbb{E}[\gamma_t^N(\varphi) \mid \mathcal{F}_{t-1}] = \widehat{Z}_{t-1}\sum_a W_{t-1}^a (K_{t,\theta}\varphi)(z_{0:t-1}^a) = \gamma_{t-1}^N(K_{t,\theta}\varphi). \tag{150}$$

Taking expectations and applying the induction hypothesis gives $\gamma_{t-1}(K_{t,\theta}\varphi) = \gamma_t(\varphi)$. The argument first applies to nonnegative functions by Tonelli's theorem; applying it to the positive and negative parts proves the signed case under absolute integrability. Set $\varphi = 1$ for the likelihood and $\varphi = h_{\theta,t}$ componentwise for its derivative, using Proposition 9 at the time prefix t. □

This proof also specifies an executable analytical recursion. A particle need carry only its state x_t^i, Jacobian $\dot{x}_t^i$, and accumulated score s_t^i, rather than an explicit array of all past disturbances. At initialization, $s_0^i = \nabla_\theta \log \rho_{0,\theta}(v_0^i)$. After choosing ancestor $a = a_t^i$, copy its state, Jacobian, and score; evaluate (133) at the newly drawn disturbance; and, under the differentiability assumptions in Assumption 1, set

$$\begin{aligned} s_t^i = s_{t-1}^a &+ \partial_\theta \log r_{t,\theta}(u_t^i \mid x_{t-1}^a) + (\dot{x}_{t-1}^a)^{\mathsf{T}} \nabla_x \log r_{t,\theta}(u_t^i \mid x_{t-1}^a) \\ &+ \partial_\theta \log g_{t,\theta}(y_t \mid x_t^i) + (\dot{x}_t^i)^{\mathsf{T}} \nabla_x \log g_{t,\theta}(y_t \mid x_t^i). \end{aligned} \tag{151}$$

Induction on this sum gives $s_t^i = h_{\theta,t}(z_{0:t}^i)$ exactly. Normalize the weights in (147) and report $\widehat{Z}_t$, $\widehat{D}_t = \widehat{Z}_t \sum_i W_t^i s_t^i$, and $\widehat{S}_t = \sum_i W_t^i s_t^i$ separately. All derivatives in this recursion hold the realised disturbances fixed; no proposal-density derivative enters the complete target score. Choosing $q_{0,\theta} = \rho_{0,\theta}$, $q_{t,\theta} = r_{t,\theta}$ and $\alpha_t = W_{t-1}$ gives the bootstrap disturbance filter, with $\omega_0^i = 1$ and $\omega_t^i = g_{t,\theta}(y_t \mid x_t^i)$. It is the simplest comparator for guided proposals.

This forward recursion preserves the support of a degenerate transition. Its particle histories can still coalesce under resampling, so it retains the long-horizon variance problem of a genealogical score estimator. Avoiding an invalid backward density resolves a support problem; it does not prove a variance bound. The next proposition separates another issue: unbiasedness of the unnormalized pair does not survive division in general. Write $\widehat{D}_N = \widehat{D}_T$, $\widehat{Z}_N = \widehat{Z}_T$, $D = \nabla_\theta Z$, and $Z = Z_T$.

Proposition 13 (Normalization creates a separate bias question)**.** *Unbiasedness of $\widehat{D}_N$ and $\widehat{Z}_N$ does not imply $\mathbb{E}[\widehat{D}_N/\widehat{Z}_N] = D/Z$. In particular, if $(\widehat{Z}_N, \widehat{D}_N)$ equals $(1/2, 1)$ or $(3/2, 1)$ with equal probability, then $\mathbb{E}\widehat{Z}_N = 1$, $\mathbb{E}\widehat{D}_N = 1$, but*

$$\mathbb{E}\left[\frac{\widehat{D}_N}{\widehat{Z}_N}\right] = \frac{1}{2}\left(2 + \frac{2}{3}\right) = \frac{4}{3} \neq 1 = \frac{D}{Z}. \tag{152}$$

Proof. The displayed two-point distribution has positive denominators and the stated first moments by direct arithmetic. Its expected ratio is the average of the two reciprocal denominators, which is $4/3$. Therefore the ratio operation does not commute with expectation in general. □

The practical output of a disturbance filter should therefore retain $\widehat{Z}_N$, $\widehat{D}_N$, and $\widehat{D}_N/\widehat{Z}_N$ as three different quantities. A stop-gradient correction for a parameter-dependent resampling law, such as the correction of Ścibior and Wood [2021, Section 3.1, equation (13)], is relevant when automatic differentiation is used to differentiate the random finite program. It is not a second term to append to the Fisher estimator in (135); that estimator already averages the complete score under the particle importance measure. The induction proves that no extra resampling-law term is missing from $\widehat{D}_t$. Appending another term changes the estimator and requires a separate expectation-preservation proof.

Proposition 14 (An exact unnormalized control variate)**.** *In the setting of Proposition 12, fix a parameter direction $v \in \mathbb{R}^p$, let $c_\theta(z)$ be a scalar, deterministic test function with $\int \pi_\theta |c_\theta| \,\mathrm{d}\lambda < \infty$, and suppose its target integral*

$$C(\theta) = \int \pi_\theta(z; y) c_\theta(z) \,\mathrm{d}\lambda(z) \tag{153}$$

is known exactly. Put $A_N = v^\mathsf{T} \widehat{D}_N$ and $B_N = \gamma_T^N(c_\theta)$. For any fixed scalar β, define

$$\widehat{D}^v_{\mathrm{cv}} = A_N - \beta(B_N - C(\theta)), \qquad \mathbb{E}\widehat{D}^v_{\mathrm{cv}} = v^\mathsf{T} \nabla_\theta Z. \tag{154}$$

If A_N, B_N have finite second moments and $\mathrm{Var}(B_N) > 0$, the variance-minimizing fixed coefficient is

$$\beta^* = \frac{\mathrm{Cov}(A_N, B_N)}{\mathrm{Var}(B_N)}. \tag{155}$$

Proof. Proposition 12 gives $\mathbb{E}A_N = v^\mathsf{T} \nabla_\theta Z$ and $\mathbb{E}B_N = C(\theta)$, so

$$\mathbb{E}[\widehat{D}^v_{\mathrm{cv}}] = v^\mathsf{T} \nabla_\theta Z - \beta C(\theta) + \beta C(\theta) = v^\mathsf{T} \nabla_\theta Z.$$

The constant does not affect variance. Its full expansion is

$$\mathrm{Var}(\widehat{D}^v_{\mathrm{cv}}) = \mathrm{Var}(A_N) - 2\beta \,\mathrm{Cov}(A_N, B_N) + \beta^2 \,\mathrm{Var}(B_N). \tag{156}$$

Differentiating this quadratic gives $-2\,\mathrm{Cov}(A_N, B_N) + 2\beta\,\mathrm{Var}(B_N) = 0$, which proves (155); the second derivative $2\,\mathrm{Var}(B_N) > 0$ proves the unique minimum. At the minimum the variance is $\mathrm{Var}(A_N) - \mathrm{Cov}(A_N, B_N)^2 / \mathrm{Var}(B_N)$. □

The proposition gives a real criterion for a control variate: its target integral must be known, or estimated with a separately proved correction. This is the same expectation-preserving role played by baselines in the score-function construction of Foerster et al. [2018, Theorem 1 and equation (4.7)], but the present proposition makes the required target integral explicit for the disturbance measure. Taking v as each coordinate vector gives the components of the derivative. For one importance draw without resampling, $A_1 = R_\theta(\mathsf{Z}) v^\mathsf{T} h_\theta(\mathsf{Z}; y)$ and $B_1 = R_\theta(\mathsf{Z}) c_\theta(\mathsf{Z})$ give the same result. A detached KDM score has no known target integral merely because it is empirically centered. A coefficient fitted on the same estimating particles can destroy unbiasedness. An independent pilot fit is admissible if, conditional on that fit, the estimating run satisfies the same identities and the resulting products are integrable. Its estimated coefficient need not attain the theoretical minimum.

To turn the control into a concrete construction, use a reference model whose unnormalized disturbance density $\widetilde{\pi}_\theta(z; y)$ and likelihood $\widetilde{Z}(\theta; y)$ are tractable. The reference must use the same disturbance coordinates and measures. A linear-Gaussian reference can have a singular state transition: its Gaussian observation likelihood is still tractable when the relevant observation covariance is positive definite. Its approximation error is corrected, rather than declared absent.

Proposition 15 (A tractable reference supplies the centering constant)**.** *Suppose $\widetilde{\pi}_\theta$ obeys differentiation under its integral, its likelihood $\widetilde{Z}$ and derivative are known exactly, and $v^\mathsf{T}\nabla_\theta\widetilde{\pi}_\theta$ vanishes outside the support of π_θ. Define, on that support,*

$$c_\theta(z) = \frac{v^\mathsf{T}\nabla_\theta\widetilde{\pi}_\theta(z;y)}{\pi_\theta(z;y)}, \qquad C(\theta) = v^\mathsf{T}\nabla_\theta\widetilde{Z}(\theta;y). \tag{157}$$

Then this C is the exact centering constant in Proposition 14. When $\widetilde{\pi}_\theta > 0$, its numerator can be computed as $\widetilde{\pi}_\theta v^\mathsf{T}\widetilde{h}_\theta$, where $\widetilde{h}_\theta = \nabla_\theta \log\widetilde{\pi}_\theta$. For independent draws $\mathsf{Z}_i \sim Q_\theta$ this gives the explicit residual estimator

$$\widehat{D}^v_{\mathrm{ref}} = \beta v^\mathsf{T}\nabla_\theta\widetilde{Z} + \frac{1}{N}\sum_{i=1}^{N}\frac{v^\mathsf{T}\nabla_\theta\pi_\theta(\mathsf{Z}_i;y) - \beta v^\mathsf{T}\nabla_\theta\widetilde{\pi}_\theta(\mathsf{Z}_i;y)}{Q_\theta(\mathsf{Z}_i)}. \tag{158}$$

Its expectation is $v^\mathsf{T}\nabla_\theta Z$. If the reference equals the target in a neighbourhood of θ, $\beta = 1$ makes this derivative estimator deterministic and exact.

Proof. The support condition and cancellation give

$$\int \pi_\theta c_\theta \,\mathrm{d}\lambda = \int v^\mathsf{T}\nabla_\theta\widetilde{\pi}_\theta \,\mathrm{d}\lambda = v^\mathsf{T}\nabla_\theta\widetilde{Z}.$$

The absolute-integrability requirement is precisely integrability of the reference derivative. Substituting this control in the importance-draw version of (154) yields (158). Integrating its residual cancels Q_θ and returns $v^\mathsf{T}\nabla_\theta Z - \beta v^\mathsf{T}\nabla_\theta\widetilde{Z}$. Adding the displayed constant proves unbiasedness. Equality of target and reference in a neighbourhood makes their derivatives equal, so at $\beta = 1$ the residual vanishes for every draw. □

The same control applies to the resampled filter through $B_N = \gamma_T^N(c_\theta)$, whose unbiasedness was proved by induction. For a factorized reference, its density ratio $\widetilde{\pi}_{\theta,t}/\pi_{\theta,t}$ and complete score can be carried along each particle's genealogy using the corresponding reference state and tangent. Merely inserting the reference score $\widetilde{h}_\theta$ omits the density ratio and is wrong relative to (157). A poor reference can make this ratio volatile or give an infinite second moment. The variance theorem therefore requires its stated moment condition and a coefficient appropriate to the complete estimator. These identities alone do not justify a particular linearization or coefficient for a DSGE model.

Exact integration is preferable when part of the original model is conditionally tractable. The next result states precisely what such integration preserves and what variance comparison it permits.

Proposition 16 (Exact conditional integration)**.** *Partition the disturbances as $z = (a, b)$ on a product reference measure, and define $\overline{\pi}_\theta(a) = \int \pi_\theta(a, b; y)\,\mathrm{d}b$. Assume differentiation under this conditional integral is valid and $\overline{\pi}_\theta(a) > 0$ on its support. Then*

$$\nabla_\theta \log\overline{\pi}_\theta(a) = \mathbb{E}_\theta[h_\theta(a, B; y) \mid a, y]. \tag{159}$$

For a single importance draw (A, B) from $Q(a, b) = Q_A(a)Q_B(b \mid a)$ covering the target support, the scalar unnormalized derivative $G = v^\mathsf{T}\nabla_\theta\pi_\theta(A, B; y)/Q(A, B)$ satisfies

$$\overline{G}(A) := \mathbb{E}_Q[G \mid A] = \frac{v^\mathsf{T}\nabla_\theta\overline{\pi}_\theta(A)}{Q_A(A)}, \qquad \mathrm{Var}_Q(\overline{G}) \le \mathrm{Var}_Q(G), \tag{160}$$

provided G has finite second moment.

Proof. Differentiate the conditional integral, use $\nabla_\theta \pi_\theta = \pi_\theta h_\theta$, and divide by $\overline{\pi}_\theta$ to obtain (159). Conditional integration under the proposal cancels Q_B:

$$\mathbb{E}_Q[G \mid A = a] = \frac{1}{Q_A(a)} \int v^\mathsf{T} \nabla_\theta \pi_\theta(a, b; y)\, \mathrm{d}b = \frac{v^\mathsf{T} \nabla_\theta \overline{\pi}_\theta(a)}{Q_A(a)}.$$

The law of total variance, $\mathrm{Var}(G) = \mathrm{Var}(\mathbb{E}[G \mid A]) + \mathbb{E}[\mathrm{Var}(G \mid A)]$, proves the inequality. □

For a genuinely conditionally linear-Gaussian block, a Kalman calculation can evaluate $\overline{\pi}_\theta$ exactly. Replacing a nonlinear block by a Gaussian approximation changes that integral; it belongs instead in the corrected reference construction above. The variance inequality compares these two coupled single-draw estimators. It does not compare complete particle filters with different proposals and future resampling decisions.

Proposition 17 (A moving-support Gaussian check)**.** *Let $U \sim \mathcal{N}(0,1)$, $X = (U, \theta U)^\mathsf{T}$ and $Y = X + \varepsilon$ with independent $\varepsilon \sim \mathcal{N}(0, I_2)$. Put $d = 2 + \theta^2$ and $b = y_1 + \theta y_2$. The observed-data score is*

$$\partial_\theta \log Z(\theta; y) = -\frac{\theta}{d} + \frac{y_2 b}{d} - \frac{\theta b^2}{d^2} = \mathbb{E}[U(y_2 - \theta U) \mid Y = y]. \tag{161}$$

Proof. The state lies on a moving line, yet its observation covariance is $\Sigma_\theta = I_2 + aa^\mathsf{T}$, $a = (1, \theta)^\mathsf{T}$. Its determinant is d and $\Sigma_\theta^{-1} = I_2 - aa^\mathsf{T}/d$. Hence

$$\log Z = -\log(2\pi) - \tfrac{1}{2}\log d - \tfrac{1}{2}(y_1^2 + y_2^2) + \frac{b^2}{2d}. \tag{162}$$

Using $d' = 2\theta$ and $b' = y_2$ gives the first expression in (161). Alternatively, completing the square in $u^2 + (y_1 - u)^2 + (y_2 - \theta u)^2$ gives $U \mid y \sim \mathcal{N}(b/d, 1/d)$. The complete disturbance score is $h_\theta(u; y) = u(y_2 - \theta u)$, so its posterior expectation is $y_2 b/d - \theta(1/d + b^2/d^2)$, the same expression. □

This example checks both the moving support and the total path derivative. It also supplies a limiting case for the control variate: using this exact Gaussian model as its own reference makes the $\beta = 1$ derivative residual zero. The normalized ratio with a still-random likelihood denominator need not be exact. The likelihood can instead be evaluated analytically in this particular example.

Corollary 1 (What the proposed method does and does not establish)**.** *Under the assumptions above, disturbance proposals and the estimator pair $(\widehat{Z}_N, \widehat{D}_N)$ preserve the declared degenerate model and give an exact unnormalized likelihood/derivative identity. They do not establish a finite-N unbiased normalized score, low variance for long horizons, global differentiability across DSGE regime switches, or an exact HMC force.*

Proof. The first statement is Proposition 9, Proposition 10, and Proposition 12. The second follows from the ratio counterexample in Proposition 13; from the fact that Assumption 1 is local and excludes unhandled regime-boundary terms. None of these expectation identities bounds variance as T grows or equates a realised noisy score with the derivative of a specified finite likelihood program. □

An exact gradient and an exact Metropolis acceptance calculation are different requirements. For a differentiable deterministic force, momentum and position updates in a symmetric leapfrog composition are triangular shears with unit Jacobian determinant; the momentum flip reverses that composition. An exact endpoint Metropolis calculation can therefore correct such a proposal even when its force is approximate. That fact does not make the force exact. A noisy score alone does not specify the auxiliary-variable proposal or acceptance law needed for a valid sampler. Those constructions and their efficiency are outside the result proved here.

9.3 Using both past and future information

Younis and Sudderth's smoother suggests a complementary proposal design. Their equation (18) draws from a mixture of forward and backward predictive approximations, allowing either set of observations to nominate a state. Their equation (19) retains a generative correction, including an auxiliary reference density; equation (20) replaces its numerator with a learned discriminative weight [Younis and Sudderth, 2024, Section 4]. The distinction matters here. A learned state predictor is not automatically a model posterior, and multiplying two posterior approximations can count prior information twice unless the appropriate reference-density factor is included.

For transition parameters, the score contains functions of (X_{t-1}, X_t), so single-time smoothing marginals alone are insufficient. A Younis-inspired smoother must supply adjacent-state or path information and the original model correction. Its most direct role is therefore an evaluable proposal for the expectation in (121), followed by model-based weighting, rather than replacement of that expectation by the paper's supervised state-prediction loss. A learned proposal would require its own target-specific training protocol and downstream validation.

10 Selective mixture gradients and control variates

The model-score recursion and the choice of mixture-gradient estimator are separate decisions. Fisher's identity does not require differentiation of every resampling draw. Mixture expectation derivatives may nevertheless be needed for proposal learning, sensitivity calculations, or a separately defined finite program. In those settings, using IWSG for every continuous movement can introduce avoidable variance.

Consider a single component $Z = \mu + h\varepsilon$, with $\varepsilon \sim \mathcal{N}(0,1)$ and fixed $h > 0$. For the affine function $\phi(z) = az + b$, the exact derivative of $\mathbb{E}\phi(Z)$ with respect to μ is a. The location-score estimator and the pathwise estimator are

$$G_{\mathrm{IW}} = (a\mu + b + ah\varepsilon)\frac{\varepsilon}{h}, \qquad G_{\mathrm{PW}} = a. \tag{163}$$

Both have mean a. Using $\mathbb{E}\varepsilon^2 = 1$, $\mathbb{E}\varepsilon^3 = 0$, and $\mathbb{E}\varepsilon^4 = 3$ gives

$$\mathrm{Var}(G_{\mathrm{IW}}) = \frac{(a\mu + b)^2}{h^2} + 2a^2, \qquad \mathrm{Var}(G_{\mathrm{PW}}) = 0. \tag{164}$$

The constant part of the integrand, which has zero true derivative, is responsible for the diverging term. Subtracting the detached baseline $\phi(\mu)$ from the score-function term produces $a\varepsilon^2$, whose variance is $2a^2$. No bias was removed; a needless source of variance was removed. Figure 1 makes the bandwidth effect explicit.

This example isolates one smooth component. Younis's argument concerns an additional difficulty: implicit mixture reparameterizations may move samples between widely separated modes and have large variance. The example does not refute that finding. It motivates treating continuous motion within a component differently from a change in component probability.

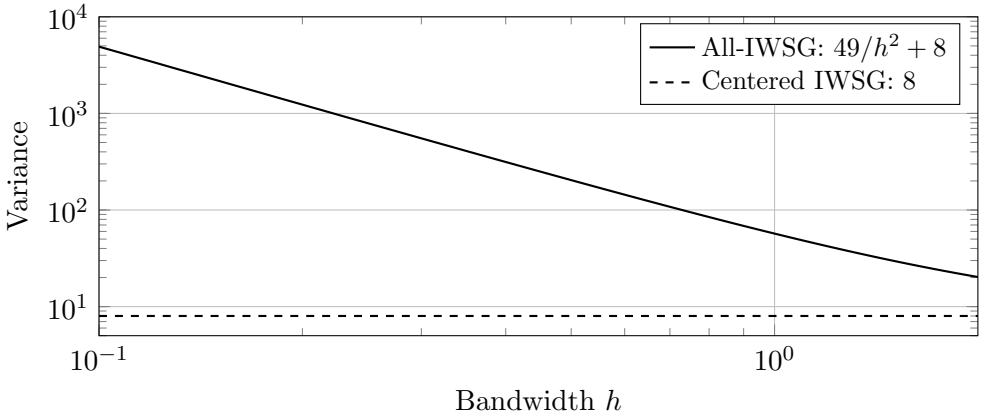


Figure 1: A narrow component can make an unbiased importance-weight derivative noisy. For $a = 2$, $\mu = 3$, and $b = 1$, the uncentered variance grows as $49/h^2 + 8$, while the centered variance stays at 8. The pathwise derivative has zero variance and therefore does not appear on the logarithmic axis. These curves are the exact calculation in (164), not filtering benchmark results.

Let $m_\theta(z) = \sum_i w_i(\theta) k_{i,\theta}(z)$, sample $I \sim \mathrm{Cat}(w)$, and represent a component sample by $Z = T_{I,\theta}(\varepsilon)$ with parameter-independent base noise. Assume positive component weights and sufficient regularity and integrability to differentiate each component expectation. Differentiating their sum gives

$$\begin{aligned} J(\theta) &= \sum_i w_i \, \mathbb{E}_\varepsilon[\phi_\theta(T_{i,\theta}(\varepsilon))], \\ \nabla_\theta J &= \sum_i (\nabla_\theta w_i) \, \mathbb{E}_\varepsilon[\phi_\theta(T_{i,\theta}(\varepsilon))] + \sum_i w_i \, \mathbb{E}_\varepsilon\left[\frac{d}{d\theta}\phi_\theta(T_{i,\theta}(\varepsilon))\right]. \end{aligned} \tag{165}$$

For a baseline c independent of the current component and noise, a resulting estimator is

$$G_{\text{hybrid}} = \frac{d}{d\theta}\phi_\theta(T_{I,\theta}(\varepsilon)) + (\phi_\theta(Z) - c)\nabla_\theta \log w_I. \tag{166}$$

The first term holds I fixed and includes both the explicit derivative of ϕ and movement of $T_{I,\theta}$. The second accounts for changing component probabilities. Its baseline has zero expectation because $\sum_i w_i \nabla \log w_i = \nabla \sum_i w_i = 0$. A component-dependent baseline does not have this cancellation unless its compensating term is retained. The baseline may depend on θ or past information, but it is held fixed within this gradient-estimation operation.

The categorical term can be averaged over its conditional component responsibilities, or components can be enumerated and sampled with a stratified allocation. Such changes have different costs and covariances with the pathwise term. Reducing the variance of one term does not automatically rank the complete estimator. A baseline fitted from the same sample also requires a justified leave-one-out or independent-fit construction; otherwise the claimed zero-mean property can fail.

The official Younis code contains an `ImportanceImplicitHybrid` branch that separates derivatives of component locations/bandwidths from derivatives of weights [Younis, 2023]. Its continuous part uses implicit mixture reparameterization. Equation (166) instead uses a componentwise pathwise derivative and is a distinct proposal. Neither the presence of that branch nor the single-Gaussian calculation establishes benefit for the full BayesFilter recursion.

Finally, the identity being estimated must remain explicit. A hybrid may estimate the same mixture expectation derivative as IWSG while differing sample by sample. Substituting it into a

fixed-anchor finite program and claiming the same derivative would require a new argument. The finite program, the expected stochastic objective, and the underlying model score cannot share a name merely because their estimators agree in a small example.

10.1 What a control variate can accomplish

For a baseline score estimate G and a control C with known mean c, define

$$G_{\mathrm{cv}} = G - \beta(C - c), \qquad \mathbb{E}G_{\mathrm{cv}} = \mathbb{E}G. \tag{167}$$

The equality requires fixed β, or suitable independence from the current estimating sample. A vector control may use a coefficient matrix. A useful correlation can reduce variance and hence MSE, but the baseline bias is retained. An independently estimated c can be used when its expectation is correct and its additional uncertainty is included. Merely naming a KDM score as the control does not provide its mean.

A useful distinction is between a KDM estimate of the model score and the score of the normalized KDM density itself. The latter does have a known center. This gives a concrete control that can be tested alongside LEDH. Its construction uses the mixture-density derivative in the IWSG identity with integrand $F = 1$; the application as an LEDH control is a local deduction, not an improvement theorem from the Younis paper.

Proposition 18 (A centered KDM density-score control)**.** *Condition on a pilot mixture with fixed positive weights π_j summing to one, means μ_j, and positive definite covariances B_j. Define*

$$m_{a,s}(z) = \sum_{j=1}^{M} \pi_j \mathcal{N}(z; \mu_j + a, e^{2s} B_j), \qquad C(z) = \nabla_{(a,s)} \log m_{a,s}(z)\big|_{a=0,s=0}. \tag{168}$$

For $Z \sim m_{0,0}$, $\mathbb{E}[C(Z) \mid \textit{pilot}] = 0$. Let a baseline vector score G use any coupling with Z that preserves these marginals. If a coefficient matrix $\boldsymbol{\beta}$ is fixed, or fitted on calibration randomness independent of the evaluation randomness conditional on the pilot, then

$$\mathbb{E}[G - \boldsymbol{\beta} C(Z) \mid \textit{pilot,calibration}] = \mathbb{E}[G \mid \textit{pilot,calibration}]. \tag{169}$$

Assume the required second moments exist when comparing variances.

Proof. Differentiating the normalized finite Gaussian mixture under its integral is valid for translation and scale in a neighborhood of the fixed positive definite covariances. Thus

$$\mathbb{E}C(Z) = \int m_{0,0}(z) \frac{\nabla_{(a,s)} m_{a,s}(z)\big|_{0,0}}{m_{0,0}(z)} \,\mathrm{d}z = \nabla_{(a,s)} \int m_{a,s}(z) \,\mathrm{d}z \bigg|_{0,0} = 0.$$

Conditioning on the independent pilot and calibration makes $\boldsymbol{\beta}$ fixed and leaves the evaluation mixture law unchanged. Linearity of conditional expectation gives the second assertion. □

Writing $r_j(z)$ for the mixture responsibilities, the translation control is $\sum_j r_j(z) B_j^{-1}(z - \mu_j)$, and the scale control is $\sum_j r_j(z)\{(z - \mu_j)^\top B_j^{-1}(z - \mu_j) - d\}$. These expressions differentiate the density at fixed z. They do not differentiate a sampled point along its generating path. For example, draw $A \sim \mathrm{Categorical}(\pi)$ independently of $\varepsilon \sim \mathcal{N}(0, I)$ and set $Z = \mu_A + \mathrm{chol}(B_A)\varepsilon$. Reusing ε in LEDH initialization preserves both marginal laws and may produce useful covariance. Averaging several such controls retains the zero center.

The coupling matters. If instead G is measurable with respect to a cloud $\mathcal{F}$ and additional KDM sampling satisfies $\mathbb{E}[C \mid \mathcal{F}] = 0$, then $\mathbb{E}[(G - \mathbb{E}G)C^\top] = \mathbb{E}[(G - \mathbb{E}G)\mathbb{E}(C^\top \mid \mathcal{F})] = 0$. Adding independent conditional mixture noise cannot reduce the variance of that already-computed cloud score through a fixed linear control. Shared estimating randomness, a correct center, and independent coefficient fitting are therefore separate requirements. Even when the resulting covariance is useful, Proposition 18 preserves the baseline's bias relative to the model score.

The earlier residual decomposition is still an exact algebraic identity:

$$L_0^N = A_h + \left(L_0^N - A_h\right), \qquad \mathrm{d}L_0^N = \mathrm{d}A_h + \mathrm{d}\left(L_0^N - A_h\right). \tag{170}$$

Dropping the residual changes the target. Computing both terms exactly, however, gives the original calculation back and supplies no automatic variance or cost reduction. The useful additional ingredient would be a cheap, accurately integrable approximation and a residual whose expectation can be estimated with less error or work.

For example, let $\widetilde{p}_\theta(x, y)$ be a tractable approximation to the joint density, with known integral $\widetilde{Z}(\theta)$. For a proposal q covering the support of the density difference,

$$Z(\theta) = \widetilde{Z}(\theta) + \mathbb{E}_q\left[\frac{p_\theta(X, y) - \widetilde{p}_\theta(X, y)}{q(X)}\right], \tag{171}$$

$$\nabla_\theta Z(\theta) = \nabla_\theta \widetilde{Z}(\theta) + \mathbb{E}_q\left[\frac{\nabla_\theta p_\theta(X, y) - \nabla_\theta \widetilde{p}_\theta(X, y)}{q(X)}\right]. \tag{172}$$

The second equality assumes fixed q and valid differentiation under the integral. If the approximation is close, the residual may be easier to integrate. The ratio of the resulting estimates still need not be an unbiased score, and a signed residual estimate can make the likelihood estimate negative. It is therefore not automatically eligible for pseudo-marginal inference. The residual formulation deserves a separate comparison if a tractable approximation can be constructed cheaply; it is not what the previous integrated KDM experiments implemented.

11 Other options and their place in the investigation

Analytic marginalization should come before additional approximation when the model permits it. Section 6 gives the one-step Gaussian example. A conditionally linear-Gaussian state block can similarly be integrated rather than sampled, provided its conditional model is retained exactly. A Gaussian approximation to a nonlinear block is a different choice and needs its own error assessment. Neither a good fit to state observations nor an appealing proposal density establishes that the downstream score is accurate.

More particles, paired random designs, and randomized quasi-Monte Carlo are important equal-compute alternatives. They can improve integration without changing the scientific target, but they cannot restore a missing derivative or correct a filtering approximation with persistent bias. The comparison must distinguish a better estimator from simply spending more work on it. Sparse mixtures, low-rank density approximations, and approximate backward sums likewise require separate bias and cost measurements; they are not algebraically equivalent to the full mixture merely because they produce finite values.

The literature audit identifies several named comparators that should be added before the score study is treated as complete. Nemeth, Fearnhead, and Mihaylova use kernel density approximation and Rao–Blackwellisation for a linear-cost score and observed-information estimator [Nemeth et al., 2016]. This is the closest published precedent for the KDM row in the present program. Its

bandwidth, kernel, and regular-transition assumptions must be carried into the experiment; a positive-bandwidth estimate is not an exact finite-particle score. For additive complete-data terms, the sequential smoother of Fearnhead, Wyncoll, and Tawn [Fearnhead et al., 2010] and the PaRIS recursion of Olsson and Westerborn [Olsson and Westerborn, 2017] provide explicit linear-cost alternatives to a generic "backward pair" label. PaRIS's number of backward draws and its mixing conditions are part of the method, not implementation details.

A qualitatively different regular-model comparator is the coupled conditional particle smoother of Jacob, Lindsten, and Schön [Jacob et al., 2020]. Rhee–Glynn debiasing of coupled CPF chains gives an unbiased smoothing expectation under meeting and moment assumptions; the authors explicitly note that Fisher's identity then yields an unbiased score when the transition density is tractable. The construction has random cost and does not make a singular or intractable DSGE transition score exact. It should be tested as a separate debiasing arm rather than folded into a variance- minimising linear combination of two biased estimates.

The continuous-likelihood econometric literature, including Malik–Pitt and DeJong et al. [Malik and Pitt, 2011, DeJong et al., 2013], is also a required comparator for the motivation behind differentiable resampling. A continuous particle likelihood can remove a resampling discontinuity while still defining a different finite-particle measure, so its derivative requires the same target and normalization audit used here. When a transition can be simulated but its density is unavailable, iterated filtering [Ionides et al., 2011] supplies a separate artificial-parameter-noise approximation with an explicit bias–variance–mixing trade-off. It is the natural simulation-only comparator for the DSGE branch, but it is not an exact finite-N score.

Twisted particle filters [Whiteley and Lee, 2014] and multilevel particle filters [Jasra et al., 2017] belong in proposal and discretization variance studies. Twisting preserves a likelihood through a change-of-measure correction, while multilevel estimators telescope coupled discretization levels; neither paper supplies a score identity. A score-level telescope or coupling must therefore be proved before either can support an unbiased-score claim. Finally, variational particle objectives and learned smoothers remain a target-boundary issue: an ELBO or a learned conditional objective cannot be reported as the observed-data score without a separate measure identity.

Bandwidth or particle-count extrapolation is another possible bias-reduction strategy, but it requires a justified expansion. A presumed h^2 smoothing bias or $1/N$ score bias cannot be used across every nonsmooth, constrained, or resampling model without evidence. Randomized telescoping is more general. For coupled levels S_l and an independent random truncation L,

$$\widehat{S}_{\mathrm{tel}} = S_0 + \sum_{l=1}^{L} \frac{S_l - S_{l-1}}{\Pr(L \geq l)}, \qquad \mathbb{E}\widehat{S}_{\mathrm{tel}} = \lim_{l\to\infty} \mathbb{E}S_l, \tag{173}$$

when absolute integrability permits exchanging sum and expectation. The expectation equality follows because each retained increment has inclusion probability $\Pr(L \geq l)$. The limit must first be shown to equal the model score. Finite expected work and finite variance impose additional conditions on the coupling and truncation tail; the displayed equality supplies neither. This makes literal unbiasedness a distinct research problem rather than a promised property of the next mixture prototype.

The original proposal-force use remains valid in a different role. A deterministic position-only field may generate a reversible, volume-preserving kick–drift–kick proposal, with acceptance computed from the declared endpoint Hamiltonian. Each kick and drift is a shear with unit Jacobian determinant. Reversing the step size reverses the sequence, and a momentum flip turns the symmetric sequence into an involution. The ordinary Metropolis ratio therefore corrects this proposal for the endpoint density; the force need not be its exact gradient. If the endpoint uses

a fixed finite-particle likelihood, it preserves that approximate target, not the exact state-space posterior. Exact pseudo-marginal inference instead needs a valid nonnegative unbiased likelihood estimate and the auxiliary randomness in the extended target. A new random score at each leapfrog step is not automatically a reversible proposal.

The BayesFilter HMC interface already distinguishes an exact score from a frozen proposal field. A future raw-coordinate field must use the typed endpoint-corrected binding with `tune_hmc_kernel`; the fixed-transport tuner is for a genuine frozen coordinate transport with its Jacobian-corrected target. This document does not change an HMC consumer or claim posterior readiness. The interface and capability registry were read to keep the proposed downstream use consistent with the actual repository.

For models with regular transition densities, a reference study can check model identity and complete initialization, establish a corrected mixture/Fisher reference, compare selective gradient estimators where needed, and then investigate scalable ancestor approximations or learned proposals. For degenerate DSGE transitions, the first unresolved task is instead a score derivation that respects the deterministic dynamics and parameter dependence of their support. Control variates, analytic integration, and increased integration effort should be evaluated alongside that sequence. Further structural DSGE implementation remains deferred while the existing repository models are checked; the support derivations below are retained because they explain which model extensions will be valid.

12 Structural support and deferred extensions

Section 9.1 explains why the proposed ordinary ancestor recursion is excluded for the degenerate DSGE target. The following examples separate that obstruction from the support of a filtering marginal and from possible coordinate-based constructions.

The relevant support belongs to the distribution being approximated. Deterministic directions in a transition conditional on one ancestor do not, by themselves, imply that the predictive or filtering marginal lies on a lower-dimensional surface. For example, consider the state $x_t = (m_t, m_{t-1})$ of an AR(2) process with independent standard normal innovations ε_t, so that

$$x_t = Ax_{t-1} + G\varepsilon_t, \qquad A = \begin{bmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} \sigma \\ 0 \end{bmatrix}, \quad [G, AG] = \begin{bmatrix} \sigma & \phi_1\sigma \\ 0 & \sigma \end{bmatrix}. \tag{174}$$

Conditional on x_{t-1}, the transition has rank-one covariance and fixes $x_{t,2} = x_{t-1,1}$. Starting from a known initial state, the covariance after two independent shocks is $GG^{\mathsf{T}} + AGG^{\mathsf{T}}A^{\mathsf{T}}$. It is positive definite when $\sigma \neq 0$, because the displayed matrix has determinant σ^2. Multiplication by a strictly positive Gaussian observation likelihood does not reduce that marginal support. Earlier uncertainty can therefore fill directions that receive no new shock at the current time.

For finitely many fixed ancestors, however, the conditional mixture in this example lies on parallel lines with different offsets $x_{t,2} = x^i_{t-1,1}$. These components do not have one common affine support. A shared tangent basis alone cannot justify evaluating all their densities at every ambient sample as though their offsets agreed. Nor does a reset of the filtering marginal necessarily retain a particular old ancestor: a constraint on the two-slice pair (x_{t-1}, x_t) must not be imposed without that conditioning. Consequently, support must be checked separately for the transition factor, the KDM insertion distribution, and the reset output.

When the distribution to be approximated does have a common affine support, a constrained construction injects noise through its stochastic coordinates:

$$u = G_\theta L_\beta \varepsilon, \qquad B_\theta = G_\theta H_\beta G_\theta^{\mathsf{T}}, \qquad \varepsilon \sim N(0, I). \tag{175}$$

For a linear support constraint $Dx = c$, the kernel must satisfy $DG_\theta = 0$. For a nonlinear manifold, a chart or retraction $x = \rho_\theta(v, \varepsilon)$ is required, together with the appropriate coordinate-space density and Jacobian. When B_θ is rank deficient, there is no ordinary full-dimensional Lebesgue density. The implementation must evaluate the mixture in innovation or chart coordinates rather than pretending that B_θ is a full SPD ambient covariance.

This constrained kernel is an extension motivated by the DSGE target, not a claim from the Younis papers. Its first tests are rank, constraint residual, and finite-difference tests for the G_θ and H_β tangents. A positive bandwidth that leaves the support is a hard failure even if its numerical score looks smoother.

The Phase 2 diagnostic implements the fixed-chart special case: a shared orthonormal basis $U \in \mathbb{R}^{d\times r}$ is frozen, the Gaussian mixture is evaluated in $U^\mathsf{T}x$ coordinates, and the orthogonal residual is checked for both values and tangents. This gives a rank-aware density with respect to the chart measure for a fixed linear support. It does not yet differentiate a parameter-dependent G_θ, chart, or retraction. Moreover, a moving support requires more than a density Jacobian. A fixed ambient sample need not remain on the support at neighboring parameter values, so the fixed-ambient IWSG ratio used in the full-rank reference may cease to exist.

An exact example makes the missing term explicit. Let $U \sim N(0,1)$, $X_\theta = (U, \theta U)$, and $Y = X_\theta + V$, where $V \sim N(0, I_2)$ is independent. The state lies on a line that rotates with θ, while the observation has covariance

$$C_\theta = \begin{bmatrix} 2 & \theta \\ \theta & 1+\theta^2 \end{bmatrix}, \qquad C'_\theta = \begin{bmatrix} 0 & 1 \\ 1 & 2\theta \end{bmatrix}. \tag{176}$$

The Gaussian score is $\frac{1}{2}\{y^\mathsf{T}C_\theta^{-1}C'_\theta C_\theta^{-1}y - \operatorname{tr}(C_\theta^{-1}C'_\theta)\}$, which equals $1/2$ at $\theta = 0$ and $y = (1,1)$. A sample $(u, 0)$ anchored at zero lies outside the perturbed line for every $u \neq 0$ and $\theta \neq 0$. The two state laws are mutually singular, so the perturbed law has no importance density with respect to the anchor law. The chart's arc-length factor is $J_\theta = \sqrt{1+\theta^2}$; its logarithmic derivative is zero at the anchor and cannot account for the nonzero observation score.

Keeping the innovation u fixed instead gives a differentiable integral, where ϕ is the standard normal density:

$$p_\theta(y) = \int \phi(u)\,\mathcal{N}(y; (u, \theta u), I_2)\,\mathrm{d}u, \tag{177}$$

$$\frac{\partial p_\theta(y)}{\partial \theta} = \int \phi(u)\,\mathcal{N}(y; (u, \theta u), I_2)u(y_2 - \theta u)\,\mathrm{d}u. \tag{178}$$

At zero, $E[U \mid Y = y] = y_1/2$, recovering the score $y_1 y_2/2$. The extra term comes from movement of the state inside the observation factor, even though the innovation density is unchanged.

For a regular chart $x = \rho_\theta(v)$ on a fixed coordinate domain, let $m_\theta(v)$ be the coordinate mixture density and $F_\theta(x)$ the downstream likelihood integrand. The corresponding differentiation rule is

$$I(\theta) = \int F_\theta(\rho_\theta(v))m_\theta(v)\,\mathrm{d}v, \tag{179}$$

$$\partial_\theta I = \int \Big[(\partial_\theta F_\theta)(x) + \nabla_x F_\theta(x)^\mathsf{T}\partial_\theta \rho_\theta(v) + F_\theta(x)\partial_\theta \log m_\theta(v)\Big]\, m_\theta(v)\,\mathrm{d}v. \tag{180}$$

This requires the usual integrability conditions for differentiating under the integral. An anchored proposal in v can provide the raw mixture ratio, but it cannot remove the chart-motion term. If a surface-density convention is used, its volume factor must be included consistently in the density

and measure. Component-dependent supports require an explicitly defined common measure or a mixed discrete and continuous formulation.

The affine specialization gives an explicit finite calculation in which all these terms can be checked. Suppose every component uses the same injective chart $x = a_\theta + G_\theta v$, where $G_\theta \in \mathbb{R}^{d\times r}$ has full column rank. Let v_j be fixed coordinate samples and let $q_0(v_j) > 0$ be their frozen proposal density. With coordinate means c_i, SPD bandwidths B_i, and normalized positive weights α_i, define

$$m_\theta(v) = \sum_i \alpha_i \mathcal{N}(v; c_i, B_i), \qquad x_j = a_\theta + G_\theta v_j, \qquad \mathrm{d}x_j = \mathrm{d}a_\theta + (\mathrm{d}G_\theta)v_j, \tag{181}$$

$$b_j = \frac{m_\theta(v_j)}{N q_0(v_j)}, \qquad \mathrm{d}\log b_j = \mathrm{d}\log m_\theta(v_j). \tag{182}$$

Here d denotes any one parameter-direction differential; multiple directions obey the same equations. All mixture components contribute. Writing $z_{ji} = B_i^{-1}(v_j - c_i)$, the complete component and mixture differentials are

$$\eta_{ji} = \frac{\mathrm{d}\alpha_i}{\alpha_i} + z_{ji}^\mathsf{T}\mathrm{d}c_i + \frac{1}{2}\left[z_{ji}^\mathsf{T}(\mathrm{d}B_i)z_{ji} - \mathrm{tr}(B_i^{-1}\mathrm{d}B_i)\right], \tag{183}$$

$$\gamma_{ji} = \frac{\alpha_i \mathcal{N}(v_j; c_i, B_i)}{m_\theta(v_j)}, \qquad \mathrm{d}\log m_\theta(v_j) = \sum_i \gamma_{ji}\eta_{ji}, \tag{184}$$

$$\mathrm{d}\gamma_{ji} = \gamma_{ji}\left(\eta_{ji} - \mathrm{d}\log m_\theta(v_j)\right). \tag{185}$$

No coordinate-sample differential appears because the v_j are fixed. The ambient state differential in (181) generally remains nonzero.

The surface-volume factor clarifies why it must not be added again to the importance ratio. Set $M = G_\theta^\mathsf{T} G_\theta$ and $J_\theta = \sqrt{\det M}$. Then

$$\mathrm{d}\log J_\theta = \tfrac{1}{2}\,\mathrm{tr}(M^{-1}\mathrm{d}M), \qquad \mathrm{d}M = (\mathrm{d}G_\theta)^\mathsf{T} G_\theta + G_\theta^\mathsf{T}\mathrm{d}G_\theta. \tag{186}$$

At the moving point x_j, the surface densities of the target and the proposal pushed through the *current* chart are respectively $m_\theta(v_j)/J_\theta$ and $q_0(v_j)/J_\theta$. Their ratio is $m_\theta(v_j)/q_0(v_j)$: the common volume cancels. The log target surface density has differential $\mathrm{d}\log m_\theta(v_j) - \mathrm{d}\log J_\theta$, while the log proposal surface density has differential $-\mathrm{d}\log J_\theta$. This construction uses a moving ambient proposal generated by fixed coordinates; it does not assert that the perturbed law has a density relative to the frozen ambient anchor law in the rotating-line counterexample.

The same chart can carry covariance marks when those marks are defined in its coordinates. For SPD coordinate marks P_i^v, define

$$\widetilde{P}_j^v = \sum_i \gamma_{ji} P_i^v, \qquad \mathrm{d}\widetilde{P}_j^v = \sum_i \left[(\mathrm{d}\gamma_{ji})P_i^v + \gamma_{ji}\mathrm{d}P_i^v\right], \tag{187}$$

$$\widetilde{P}_j^x = G_\theta \widetilde{P}_j^v G_\theta^\mathsf{T}, \tag{188}$$

$$\mathrm{d}\widetilde{P}_j^x = (\mathrm{d}G_\theta)\widetilde{P}_j^v G_\theta^\mathsf{T} + G_\theta(\mathrm{d}\widetilde{P}_j^v)G_\theta^\mathsf{T} + G_\theta \widetilde{P}_j^v (\mathrm{d}G_\theta)^\mathsf{T}. \tag{189}$$

The ambient mark is allowed to be singular. These equations specify the transport of marks defined in this common chart; they do not infer a full DSGE UKF covariance lifecycle from a singular process covariance.

To expose the state-motion term at a consumer, take an observation $y \mid x \sim \mathcal{N}(H_\theta x + o_\theta, R_\theta)$, with y fixed and R_θ SPD. The finite log normalizer and its total differential are

$$\widehat{L}(\theta) = \log \sum_j b_j g_\theta(y \mid x_j), \qquad \pi_j = \frac{b_j g_\theta(y \mid x_j)}{\sum_\ell b_\ell g_\theta(y \mid x_\ell)}, \tag{190}$$

$$\mathrm{d}\widehat{L} = \sum_j \pi_j \big[\mathrm{d}\log m_\theta(v_j) + \mathrm{d}\log g_\theta(y \mid x_j)\big]. \tag{191}$$

There is no subtraction of $\mathrm{d}\log \sum_j b_j$; that would differentiate a self-normalized finite value. With $e_j = y - H_\theta x_j - o_\theta$ and $s_j = R_\theta^{-1} e_j$, the observation term is explicitly

$$\mathrm{d}\log g_\theta(y \mid x_j) = s_j^{\mathsf{T}} \big[(\mathrm{d}H_\theta)x_j + H_\theta \mathrm{d}x_j + \mathrm{d}o_\theta\big] + \tfrac{1}{2}\Big[s_j^{\mathsf{T}}(\mathrm{d}R_\theta)s_j - \mathrm{tr}(R_\theta^{-1}\mathrm{d}R_\theta)\Big]. \tag{192}$$

The term $H_\theta \mathrm{d}x_j$ is exactly what a zero outgoing state tangent would omit. Differentiating this finite sum needs no exchange of derivative and integral. Relating it to a limiting likelihood still needs an importance sampling argument and appropriate integrability conditions. Its derivative is neither the unchanged atom-program derivative nor, at finite N, the exact model score.

Support preservation by the reset is a separate question. On the parabola $x = (u, u^2)$, a convex barycentre with weights λ_i obeys

$$\bar{x}_2 - \bar{x}_1^2 = \sum_i \lambda_i u_i^2 - \left(\sum_i \lambda_i u_i\right)^2 = \mathrm{Var}_\lambda(u). \tag{193}$$

Thus even exact barycentric averaging leaves this curve whenever the weighted variance is positive. Contract–E adds residual injection and affine moment restoration, so barycentric failure alone does not prove its final residual; the actual reset must be evaluated. On a common affine subspace, ambient ridge and residual injection can likewise introduce normal-direction variation. Projecting the result or performing the reset in chart coordinates changes the specified finite reset and requires its own derivation; it cannot silently count as the existing Contract–E program.

The existing subspace kernel rejects points outside the rotating line, and the independent observation-score check recovers $1/2$ and its finite difference. The affine equations above now have a diagnostic implementation with a Gaussian observation consumer. Its two-direction tests perturb each mixture, chart, mark and observation input separately and jointly. The largest absolute difference from an independent central-difference calculation is 6.11×10^{-11} on these fixtures. An executable wiring check follows that consumer to the existing complete mixture evaluator. Deliberately zeroing its outgoing state tangent changes the computed normalizer derivative while leaving the value unchanged.

The reset probes are untuned CPU/float64 references, not measurements of production filtering accuracy. They use eight particles, Sinkhorn $\epsilon = 0.2$, 20 iterations plus five balance iterations, and ridge 10^{-5}, followed by the actual diagonal and pairwise correction with both caps enabled. On the line $x_2 = 0$, the maximum absolute constraint residual is 0.00374 after Contract–E and 6.52×10^{-7} after correction. On $x_2 = x_1^2$, the corresponding residuals are 0.918 and 1.187. Both correction calls return numerical validity. That flag therefore does not certify either of these support constraints. These counterexamples concern prescribed common supports; they do not impose an old ancestor's conditional support on every filtering marginal.

The shared finite LEDH executor still factors a full-rank process covariance and uses full-rank Gaussian PF–PF factors. A target-specific innovation-coordinate flow, reset, density recurrence, and their analytical differentials must be specified together before an end-to-end DSGE implementation

can be claimed. Its initial state and covariance tangents are also set to zero: a parameter-dependent initial law requires explicit derivatives at that entry point, whereas the completed reference campaigns held those inputs fixed.

13 Implementation history and current replication status

13.1 What the October source closure establishes

The 14 October replication result closes the bounded controlled-SMC and released 25-particle bearings MDPF/A-MDPF source checks. Its 283 checks concern evidence, provenance and wiring; they are not 283 new numerical experiments. The result also records agreement of eight numerical runtime files with their saved evidence. Full-paper reproduction, population-level trained equivalence, original-iAPF replication and the LEDH enhancement experiments remain outside that closure. The immediate next task is to verify the integration and target identities before comparing proposal quality or fixed-law score estimators. No new LEDH improvement is established by the source closure itself.

13.2 The September LEDH–KDM implementation snapshot

The remainder of this section describes the September snapshots, including their then-observed gaps. These call-chain claims have not been revalidated against every current consumer endpoint in this document audit. The reported numerical results belong to research commit `804616e3`; the comparison checkout inspected then was `5cc59cfa`. The analytical function and legacy batch wrapper are the two functions named in Section 2, with the source-route initialization in `bayesfilter/highdim/ledh_pfpf_genut_initial_rqmc_tf.py`. The existing prior-weight sidecar captures a corrected pre-observation cloud, but it does not implement the Younis IWSG mixture gradient through the integrated LEDH–OT–GenUT reset. It must not be relabelled as that method.

The registry call-chain repair is complete. Phase 1 is now implemented as the diagnostic module `bayesfilter/highdim/ledh_younis_kdm_tf.py`. Its route ID is

`ledh_younis_kdm_algebra_diagnostic_v1`

Its global and conditional mixture kernels expose the all-pairs density and complete cloud, weight, and covariance tangents. The source-matched fixed-proposal IWSG kernel has no sample-location tangent and keeps the proposal density at its supplied anchor. The conditional kernel is labelled as a proposal-density evaluator, not a model-score estimator. The separate anchored `MODEL-IS` PF–PF kernel contains every value term in (79)–(80) and requires an explicit per-particle certificate that the forward map is invertible; a finite log determinant alone is not accepted as that certificate. Its tangent excludes proposal and selection derivatives because their sampling law is frozen at the anchor. The module is diagnostic-only and is not in the canonical registry or call chain.

The bounded Phase 1 test suite passes its complete-tangent, rank, anchor, proposal, and XLA-smoke checks. Phase 2A adds

- the controlled one-step linear-Gaussian normalizer `make_linear_gaussian_kdm_normalizer_kernel`, whose positive branch is `KDM-FINITE` and whose explicit zero branch is `ATOM-FINITE`; and
- an opt-in `return_trace=True` surface on the analytical endpoint, used to build the sidecar from the actual post-flow cloud and weights under the Contract–E/dual-cap settings.

The Phase 2A finite-difference and endpoint tests pass, and the Phase 2B fixed-chart support tests pass. Phase 3 now provides `canonical_linear_gaussian_kdm_auxiliary`. It reads the actual analytical trace, reconstructs the zero-bandwidth value and score from (14), and reports a separately labelled positive-bandwidth functional without feeding it into the reset. In particular, it uses $\bar{w}_t$, not w_t^+; using w_t^+ would apply the current observation twice.

Phase 4A is now implemented in `bayesfilter/highdim/ledh_younis_kdm_integrated_tf.py`. The route `ledh_younis_kdm_integrated_observation_weighting_v1` calls the same internal finite-program executor as the registered canonical wrapper and substitutes only the observation factor. The code implements (43), feeds the resulting weights and tangents through Contract–E and both correction caps, and returns the actual multi-step trace. Its fixed-shape factory defaults to XLA compilation.

The implementation audit also repaired three pre-existing total-derivative gaps in the shared analytical executor: generic Gaussian transition and observation fallbacks now include direct covariance-density tangents; a model may supply the total tangent of its observation Jacobian; and every resulting dH term enters the LEDH flow. The flow log determinant now uses an XLA-compatible QR/triangular-solve identity rather than an unsupported matrix determinant/inverse pair. Rebuilt-model finite differences test these terms.

Phase 4A is a correct total derivative of its declared `KDM-FINITE` scalar; it is wrong if described as the derivative of `ATOM-FINITE`. The current one-cell experiment gives no evidence that it reduces model-score error, and the all-bandwidth inspection indicates that its small variance reduction is offset by additional bias. The complete rows, manifest, and source hashes are recorded under `docs/benchmarks/artifacts/ledh_younis_kdm_phase4_20260908/campaign-attempt05-n32t5-all-rho-f32-current/`.

Phase 4B is no longer blocked by an unspecified mixture or ancestry rule: Equations (53)–(63) define the source-matched raw mixture ratio, the BayesFilter covariance marks, the insertion point after Contract–E, and the two-pass anchor/replay rule. The exact N-by-N operation is implemented in `bayesfilter/highdim/ledh_younis_kdm_resampling_tf.py`. Its anchor and replay endpoints call the shared canonical executor after Contract–E and both correction caps, enforce one common full-rank Gaussian bandwidth at each time, and expose both raw and self-normalized ratio diagnostics with complete tangents. The repaired route carries raw ratios divided by N, not self-normalized ratios, into the next PF–PF normalizer. The responsibility-mean and fixed-label covariance-mark policies are separate, explicit routes. One-at-a-time and joint finite differences, exact anchor replay, canonical no-hook parity, and isolated two-step state/weight/covariance wiring tests pass. Bounded float64 and float32/no-TF32 GPU/XLA smokes of the earlier self-normalized route are superseded for derivative certification. On 9 September, four fresh smokes of the repaired raw-IWSG program passed for both precisions and both covariance mark policies at $D = 2, N = 8, T = 2$. These checks establish implementation consistency for `RESKDM-IWSG-FINITE`. The subsequent matrix campaign supplies the separate negative model-score result reported above. The conditional `MODEL-IS`

proposal remains an extension and is not a substitute for this combined program. The canonical trace now retains, rather than discards, higher-moment validity, the off-diagonal target mask, pairwise pre/post-cap displacement, pairwise cap scale, and coordinate-cap activity. The new matrix Kalman recursion and fixed-stream bootstrap comparator pass analytical-versus-finite-difference tests. A terminal audit reproduced both campaign summaries from the raw rows and verified source hashes, split counts, validity flags, and complete correction diagnostics. Its receipts are retained under `docs/benchmarks/artifacts/ledh_younis_kdm_phase4b_20260909/campaign01/`. The campaign completed, but its planned power requirement was not met. The affine-chart diagnostic in `bayesfilter/highdim/ledh_younis_kdm_affine_chart_reference_tf.py` implements (181)–(192), including two-direction and heterogeneous-coordinate-mark tests. All 35 focused CPU/float64 reference checks pass. It is not wired into the multi-step LEDH executor and has not been validated on GPU/XLA. Further DSGE implementation is deferred. The five-adapter checks below establish a useful starting point for current model support. Before those models support any new score-quality conclusion, their initial laws and precise observation and transition targets must be resolved and the complete consuming filter tested under fresh model-specific calibration. An existing model that the current executor cannot represent must remain explicitly untested; it must not be replaced by a different full-rank model to fill a results table.

The first cross-model audit exposes why component fidelity is not enough. The generalized-SV adapter evaluates the actual heteroskedastic likelihood, but its flow also depends on the process covariance and on the observation map. Write its parameters as $\theta = (a_s, a_h, \log\sigma_s, \log\sigma_h, \log\beta)$ and perturb them in direction v. The process covariance and flow observation map obey

$$Q = \operatorname{diag}(\sigma_s^2, \sigma_h^2), \qquad \mathrm{d}Q = \operatorname{diag}(2\sigma_s^2 v_3, 2\sigma_h^2 v_4), \tag{194}$$

$$H = [\beta, 0], \qquad \mathrm{d}H = [\beta v_5, 0], \tag{195}$$

$$\mathrm{d}(Hx) = H\,\mathrm{d}x + (\mathrm{d}H)x. \tag{196}$$

Here parameter components are numbered from one. The adapter previously omitted $\mathrm{d}Q$, $\mathrm{d}H$, and the second term in the last line. Correct differentiation of the likelihood at a supplied state cannot restore those missing derivatives of the state produced by the flow. The KSC adapter had the same problem with its flow observation offset: the differential of $h + \mu_{\mathrm{mix}} + 2\log\beta$ is $\mathrm{d}h + 2\mathrm{d}\log\beta$, not just $\mathrm{d}h$. Finite differences that rebuild both adapters at the perturbed parameters expose these omissions. They are implementation errors relative to the total-derivative claim, not evidence against KDM. The KSC mixture must also remain distinct from exact transformed SV: replacing log-chi-square noise by seven Gaussian components changes the observation density, so an observation-transform Jacobian does not imply equal scores.

After these repairs and a graph-construction repair for Austria SIR, the first cross-model batch passes 39 checks. All five existing adapters—diagonal LGSSM, native generalized SV, KSC, predator–prey, and the current Austria SIR recurrence—pass coordinate-by-coordinate and mixed-direction finite differences through the two-step canonical and raw-IWSG replay endpoints. The largest scaled discrepancy is 2.50×10^{-8}. These are CPU/float64 reference tests with fixed initial clouds and covariances. The executor still initializes their tangents to zero, whereas a stationary SV initial law depends on its persistence and innovation scale. Its total model-score derivative therefore needs additional initialization terms. The continuous SIR recurrence also omits the susceptible clipping in the repository simulator and its latent-preclip representation. Its derivative test does not establish either model identity. Both discrepancies must be resolved before those models enter the score-accuracy comparisons; the remaining repository models are still untested by this batch.

Fresh Phase 4A endpoint calibration after the covariance-carry repair passes for float64 and for float32 without TF32. Its float32 TF32 arm still fails the declared graph/XLA and atom-identity

tolerances, so TF32 remains unadmitted for that diagnostic route. Phase 4B has so far been checked only in float64 and float32 without TF32. The batch implementation has changed since the earlier limitation was written. In the inspected main checkout, `ledh_canonical_batch_fused_tf.py` now calls the single-cloud analytical executor and forwards Contract–E, higher-moment, pairwise-cap, and coordinate-cap settings. It uses nested `tf.map_fn` calls over rows and parameter directions. This is a row-mapped scalar implementation; despite the filename and its description, it does not satisfy the repository's requirement for a batch-native NeuTra training target. The older `ledh_canonical_batch_tf.py` wrapper also maps rows and does not itself forward a Contract–E reset selection, so its default invocation remains the diagnostic reset-free slice.

The shared main-checkout score executor now reaches `batched_higher_moment_shape_jvp` in `ledh_unified_correction_tf.py` through a batch-axis adapter. This source change means the old experiment hashes cannot certify the current shared numerics. Source wiring was inspected for this revision; numerical parity of that later dependency chain was not rerun. The integrated observation and raw-IWSG source files match the research snapshot, but their shared dependency has changed. Implementation reuse is therefore distinct from renewed numerical certification.

The new marginal proposal, ancestor-averaged Fisher recursion, and selective component gradient in Sections 6–10 are derived proposals. They are not integrated, tested runtime candidates in either snapshot. The existing anchored `MODEL-IS` kernel is a one-step diagnostic and supplies only part of that future implementation. A useful model-score estimator could differ from the derivative of (10); its value would be established against the underlying model score and through its own implementation checks. The present results establish neither HMC readiness nor a change to the canonical default.

13.3 Checks on the existing finite-program endpoints

The existing TensorFlow diagnostics reconstruct the Younis mixture algebra with analytical tangents. The proposal-corrected LEDH arm is a separate extension, not a source theorem. The evaluator must expose cloud, weight, and bandwidth derivatives and provide a small all-pairs reference. Any integrated claim requires the consuming endpoint to call the evaluator with compatible shapes, dtypes, and derivatives. The proposed model-score routes may have their own endpoints; a standalone density helper does not establish integration with any complete filter.

The finite-program checks have the following distinct purposes:

1. For Phase 4A, at zero bandwidth, value and score match the canonical fixed-stream program within the declared numerical tolerance.

2. The Phase 4A zero-bandwidth branch is an atom calculation; no Cholesky factor or inverse of a zero covariance is used to manufacture the limit. Phase 4B makes no zero-bandwidth parity claim.

3. At positive bandwidth, the result records $\overline{L}_h^N - L_0^N$ and is not labelled value-preserving.

4. Finite differences of the declared L_h^N agree with its complete analytical tangent when state, observation matrix, process and observation covariances, bandwidth, and model parameters are perturbed separately and jointly.

5. Before any future degenerate DSGE extension is admitted, its fixture must have zero support violation and a correct rank-aware density calculation. That extension is deferred while existing repository models are tested.

6. Any route claiming linear cost needs a scaling check with its stated assumptions; an all-pairs N^2 operation is incompatible with that claim. The all-pairs implementation remains diagnostic.

7. On a linear-Gaussian model, score error is compared with the Kalman score over paired paths and independent fixed streams. Bias and variance are reported separately.

8. For a `MODEL-IS` arm, the proposal-density identity is checked separately from the post-reset finite value; the complete mixture denominator, support, and forward Jacobian must all be present.

9. For Phase 4B, every anchor sample is replayed exactly, its forward importance ratio is one, every component contributes to the marginalized density, and the N^2 responsibility and covariance-mark tangents agree with finite differences.

10. The sequential Phase 4B endpoint is tested at $T \geq 2$ so that the post-reset KDM state, weight, and covariance marks demonstrably affect a later normalizer. A one-step density helper is not integration evidence.

11. The next PF–PF step consumes the raw IWSG log weights and their uncentered tangents from (64); executable ablations must detect either one being discarded or prematurely normalized.

12. The score-quality fixture has at least two state dimensions. Its executed trace must contain a nonzero off-diagonal pairwise target mask, a finite nonzero pairwise correction displacement, and both pairwise and coordinate-cap diagnostics. Merely setting nonzero options in a scalar model does not execute the pairwise algorithm.

14 The next discriminating comparison

For models with regular transition densities, the reference question is whether model-corrected mixture proposals and ancestor averaging reduce model-score error at equal compute. This is a restricted comparison, not an implementation plan for degenerate DSGE transitions. A method intended for that target must first pass the conditional-support analysis in Section 9.1; more accurate results on full-rank Gaussian models cannot remove that obstruction. Proposal quality, score construction, and mixture-gradient variance must be separated. Otherwise a successful run could be attributed to the wrong mechanism, just as a passing derivative check previously obscured a changed scalar.

The smallest useful sequence begins with three calculations. First, compare (71) and (78) with direct Gaussian integration on a fixed incoming cloud. Second, compare the complete-data and backward-recursion scores with exact enumeration on a small finite-state model, including a parameter-dependent initial law. Third, compare IWSG, centered IWSG, and (166) for the same declared mixture expectation. These calculations test identities and failure mechanisms; their particle counts need only be large enough to exercise the operation being tested. They are not score-quality campaigns or reasons to select a production particle count.

The subsequent accuracy study should use a linear-Gaussian model with nondegenerate transitions and an independent Kalman score, a short Gaussian-mixture model with evaluable transition densities whose modes can be enumerated exactly, and a scalar nonlinear model with a regular transition density and a numerically converged quadrature score. Kalman checks the basic integration and complete derivatives. The mixture model tests the separated modes that motivate Younis's method. The nonlinear model tests how proposal approximation affects the result. A one-step or

unimodal success cannot establish the latter two properties. Models with unresolved initialization or support identity must be repaired before their score results are interpreted.

For each model, change one mechanism at a time: ordinary ancestor sampling versus marginal mixture weights; a genealogical complete-data score versus the full ancestor average; and, where an expectation derivative is actually used, all-IWSG versus the selective estimator. Include the enhanced combination only after the component comparisons. The exact full-mixture reference remains necessary when evaluating a later sparse or sampled approximation.

The cheap alternatives are constructed from the model before claim data are examined. A bootstrap PF proposes from the physical transition and uses a proper complete-data score estimator. A locally adapted PF uses the current observation to guide each ancestor's proposal and keeps the exact importance weight. In nearly linear regimes, EKF and UKF approximate-likelihood scores test whether inexpensive Gaussian propagation already suffices; their derivatives must include the full declared Gaussian approximation. A fixed-setting LEDH filter with $\epsilon = 1$ and five Sinkhorn iterations supplies the current project development comparator. All other numerical controls must be declared and held fixed within a comparison. This temporary owner-selected setting does not replace the scope-specific tuning required before a confirmatory or default-readiness claim. The exact Kalman, mixture-enumeration, or quadrature solution supplies the oracle where it is available; an approximate reference must carry its own uncertainty. Neither a weak fixed-ancestry derivative nor this untuned LEDH baseline should stand in for all conventional particle-score methods.

Evaluate these comparisons conditionally. Weak and strong observations probe proposal concentration; overlapping and separated modes probe mixture gradients; low and high persistence probe loss of ancestor diversity; short and long horizons probe accumulation; near-zero and large score components probe the error scale. These situations must be defined from the model and research question before using validation data. Failure against a cheap alternative in a salient situation rejects promotion under the repository's sanity-check rule, even if the method wins an unconditional average. The adversaries are falsification checks, not extra tuning targets.

For dataset d and independent filter randomization r, denote the estimate by $\widehat{s}_{d,r}$ and the oracle by s_d. Replication within each fixed dataset estimates b_d and V_d from (3); the distribution over datasets measures generalization. Pair methods within datasets and respect this nesting in confidence intervals. A path bootstrap with only one filter realization per path cannot estimate each dataset's conditional bias. Freeze parameter coordinate scales before comparison, because a relative error divided by a nearly zero oracle score is unstable. Change N and T independently, and report both equal-particle and equal-measured-compute comparisons.

Before a serious run, choose a practically meaningful reduction δ and test a predeclared upper confidence bound for

$$\Delta = \mathbb{E}\big[\|\widehat{s}_{\text{candidate}} - s\|_D^2 - (1-\delta)\|\widehat{s}_{\text{comparator}} - s\|_D^2\big], \qquad \|v\|_D^2 = v^{\mathsf{T}} D v, \tag{197}$$

where D is the fixed coordinate scaling and the expectation follows the specified sampling of datasets and filter randomizations. The upper bound must be below zero, subject to validity and conditional-comparison vetoes. The old ten percent criterion is not automatically a justified threshold for every new model. Its practical meaning, replication requirement, multiplicity handling, and total compute budget must be stated for the new study. A power pilot that requires more replications than the budget permits leads to an inconclusive study, not a forced ranking among viable candidates.

All material numerical choices need a provenance and failure explanation. The physical transition covariance comes from the model; proposal bandwidth, defensive-mixture fraction, covariance marks, chart, regularization, and ancestor-sampling precision are numerical hypotheses. The earliest checks should expose wrong support, missing proposal mass, large importance ratios, or a gradient

dominated by inverse bandwidth. Calibration and validation must be disjoint for each bound model/horizon/particle/backend scope. The initial law is part of that scope. Retaining an old setting for comparability does not make it a tuned default, and a failed holdout cannot become the next calibration set.

Numerical validity is assessed before score ranking. Nonfinite values, incorrect density ratios, support violations, missing initial-law terms, an invalid reference, or an omitted sampling-law derivative invalidate the affected comparison. ESS, mode overlap, cap activity, tangent norms, bandwidth, memory, and runtime explain its behavior; they do not replace the model-score criterion. A valid candidate that loses is a candidate rejection. A later prespecified repair may continue unless the target, evidence, or total budget has become invalid.

BayesFilter implementation remains TensorFlow/TFP with analytical recursive scores. Autodiff and finite differences are independent diagnostics. The production direction remains GPU/XLA with the declared memory-growth and transport-chunk policies; the earlier float64 and no-TF32 tests are reference exceptions. New mixture evaluation must expose its pair count and memory behavior. Batch wrappers must preserve the mathematical calculation and, for NeuTra training, actually evaluate a batch rather than apply a scalar target independently to each row. None of the proposed new calculations is promoted by this document.

For the regular-transition comparison, the next evidence needed is correct model-density weights and complete initialization, an independently checked score recursion, and a powered comparison against simple alternatives. The strongest alternative explanation for any improvement is better proposal adaptation or more effective computation, rather than the gradient identity. The component comparisons distinguish those explanations. Loss to a tuned conventional score estimator at equal compute would overturn the practical priority of the new combination without invalidating its algebra.

A Source and experiment locations

The original Section 3.6 manuscript is `docs/papers/section_3_6_prior_weight_sidecar/section_3_6_prior_weight_sidecar.tex` in the retained worktree `.claude/worktrees/section-3-6-prior-weight-sidecar/`. The present manuscript is `docs/papers/ledh_younis_kdm_score/ledh_younis_kdm_score.tex`. The protected source for the September revision was recovered from commit `804616e320d940a5ea288ba25cc43bd21d267abb` on `kdm-total-score-continuation-20260909`. Its comparison checkout was `5cc59cfa26b8dec2b5a9e46497ef5639f2bfdb97`.

The October status update uses `docs/plans/controlled-smc-kdm-source-closure-result-20261014.md` and `docs/plans/ledh-controlled-smc-kdm-integration-checkpoint-20261014.md` from the recovered main integration worktree. The linked manuscript was audited in the separate shared `surrogate-hmc` checkout; the manuscript at the same path in that main worktree was older and was not silently replaced. Audit scope, exact checkout locations, the protected baseline and build checks are recorded in `docs/plans/ledh-kdm-manuscript-audit-20261014.md`. The controlled-SMC citation was checked against the locally retained published paper, Sections 4.2–4.3, and the source-closure record; the local author source is under `.localresources/code/heng-controlled-smc/`.

Phase 4A records are under `docs/benchmarks/artifacts/ledh_younis_kdm_phase4_20260908/`; the corrected all-bandwidth diagnostic is `campaign-attempt05-n32t5-all-rho-f32-current/`. Phase 4B records are under `docs/benchmarks/artifacts/ledh_younis_kdm_phase4b_20260909/campaign01/`. These paths describe the research snapshot and retained research worktree; their absence from another checkout is not evidence that the experiment did not run. The recovered

numerical evidence and primary-source ledger are in `docs/plans/artifacts/younis-score-recovery-20260911/`. The accompanying scientific analysis is `docs/plans/younis-model-score-analysis-2026-09-11.md`.

The derivations imported from the literature were checked at the following locations: Younis and Sudderth (2023), Section 4, equations (14)–(15), and the official resampling code; Younis and Sudderth (2024), Sections 2.3 and 4, including the generative and learned smoother weights; Lai et al. (2022), Algorithm 2, Theorem 2, equations (9)–(13), and the author's full-mixture importance weights; Poyiadjis et al. (2011), Section 2.2, Algorithm 2, equations (20) and (22); Ścibior and Wood [2021, Section 4.2, equations (24)–(25)]; and Olsson and Westerborn, Section 3.1, Algorithm 2, and Section 3.2.4, Theorem 10 with its mixing assumptions. Local paper copies are retained under `.localresources/papers/`. These sources support the component constructions; their combination with the proposed BayesFilter model-score comparison remains to be evaluated.

The September 14 literature-gap audit added local technical copies and source checks for Nemeth et al. (kernel-density/Rao–Blackwell score), Fearnhead et al. (sequential smoothing), Jacob et al. (coupled conditional-particle debiasing), Ionides et al. (iterated filtering), Whiteley and Lee (twisted particle filters), and Jasra et al. (multilevel particle filters). Their algorithms and assumptions are summarized in `docs/plans/artifacts/particle-filter-score-literature-gap-audit-20260914/`. The audit treats these methods as regular-model comparators or proposal and discretization controls unless a separate target identity is proved. It found no generic theorem that makes ambient Gaussian KDM or PaRIS valid for a deterministic or singular DSGE transition.

The revision's protected baseline, source archive, equation-preservation review, exact arithmetic examples, implementation wiring inspection, and PDF build records are retained under `docs/plans/artifacts/younis-manuscript-update-20260911/`. This revision added analysis and exposition, and ran document and reference checks. It launched no new particle-filter experiment.

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